

BFS Traversal

Step 1: Start at E

Complete

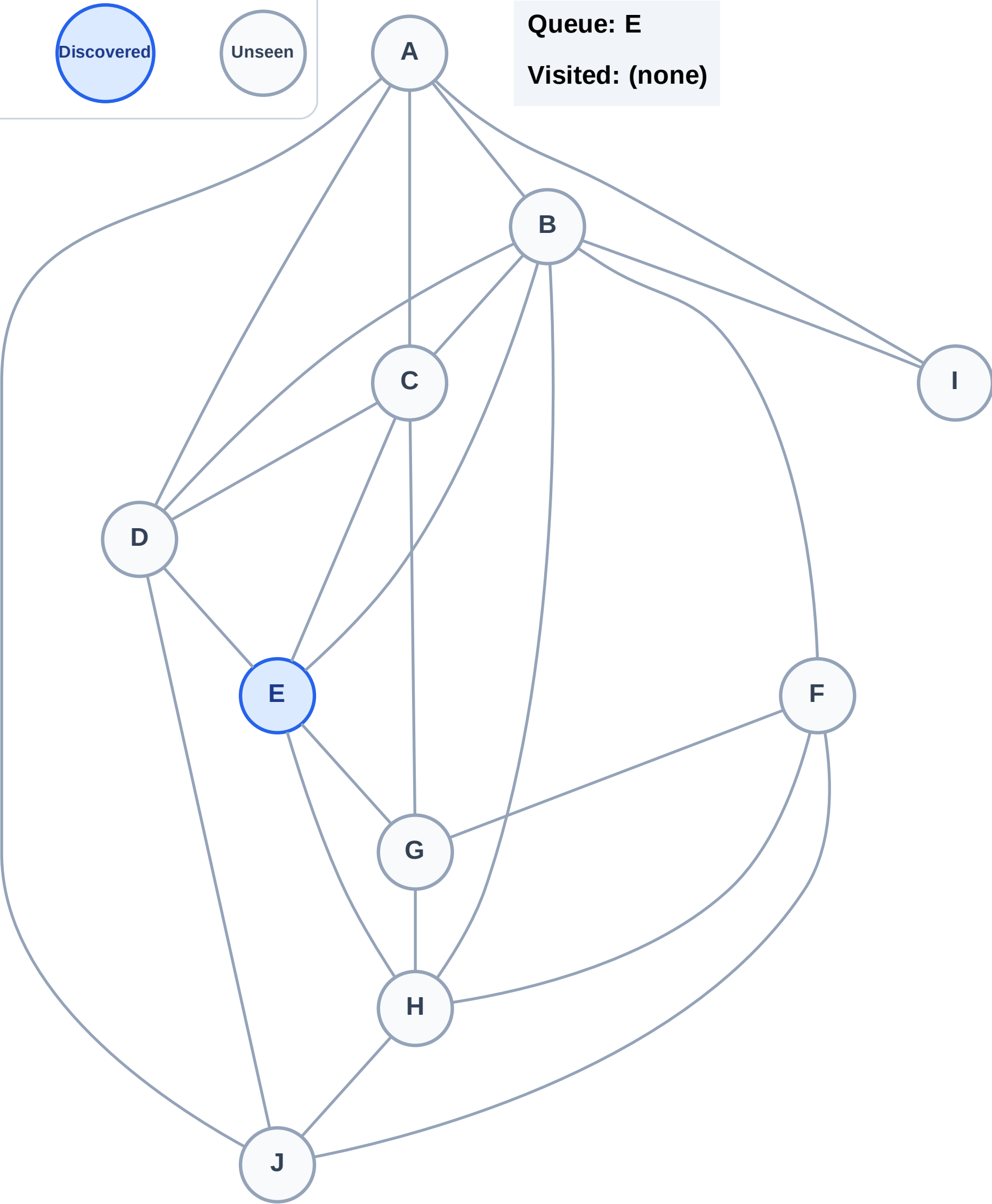
Processing

Discovered

Unseen

Queue: E

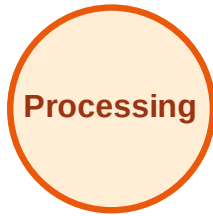
Visited: (none)



BFS Traversal

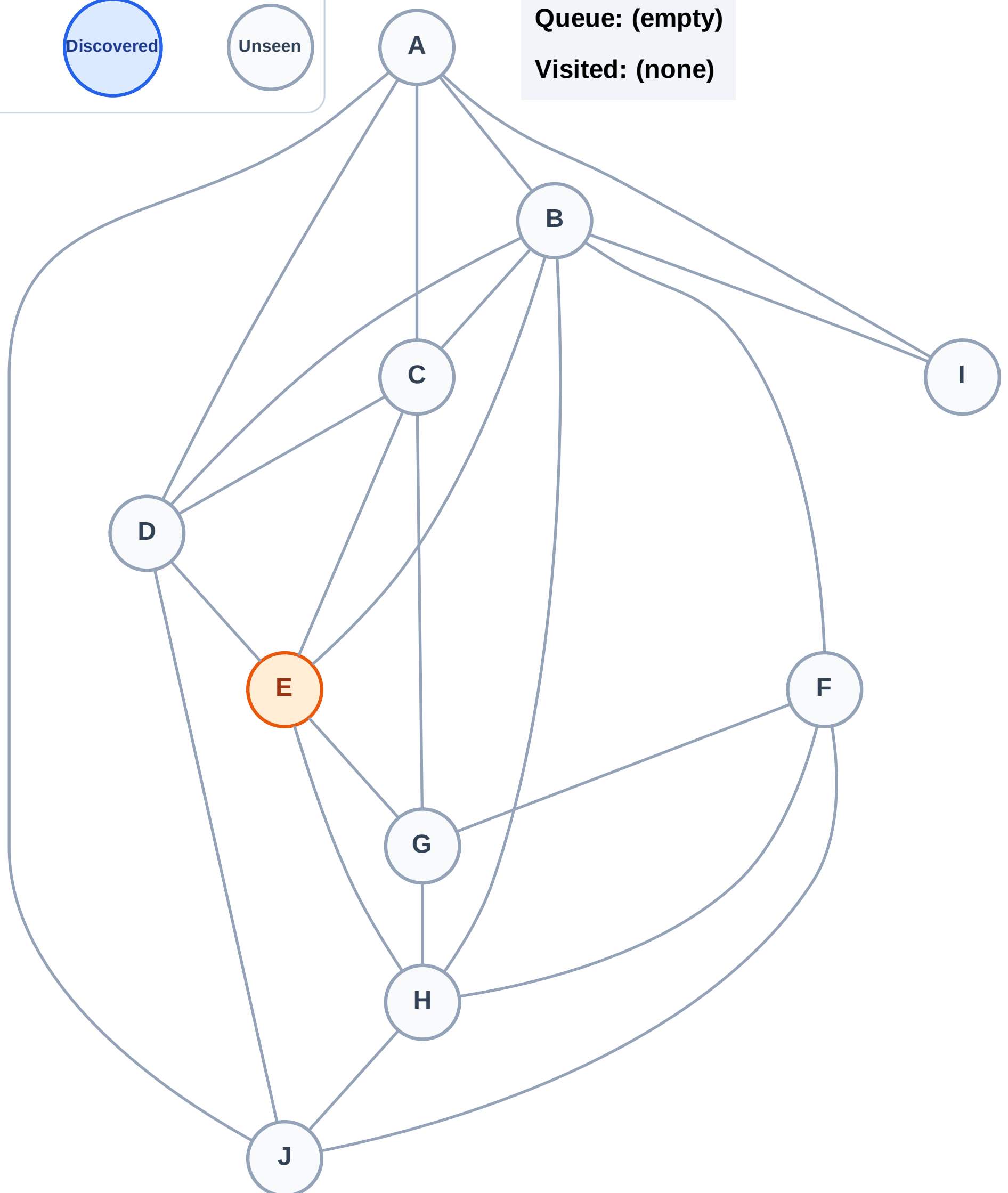
Step 2: Process node E

Legend



Queue: (empty)

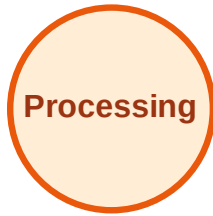
Visited: (none)



BFS Traversal

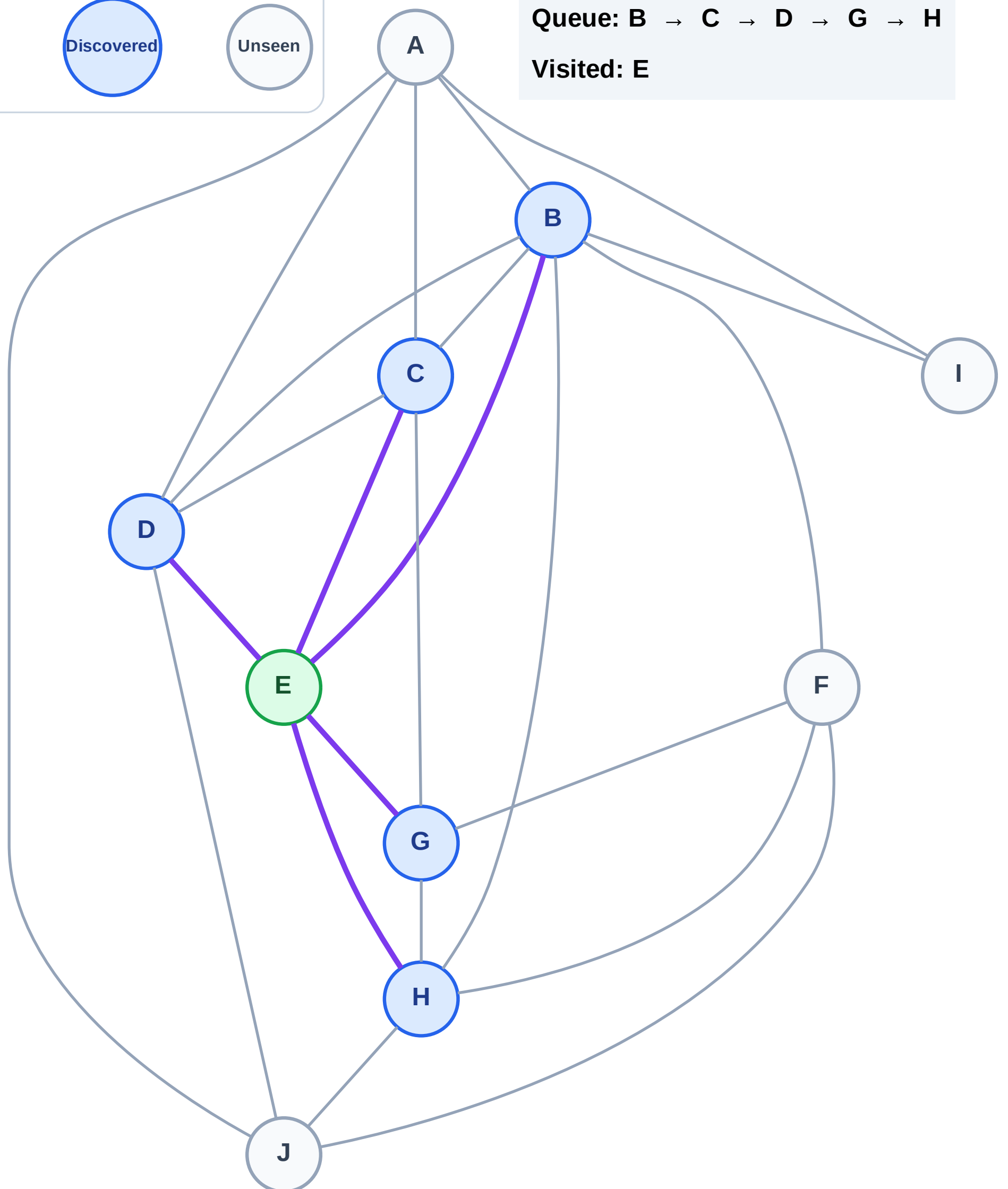
Step 3: Discover B, C, D, G, H from E

Legend



Queue: B → C → D → G → H

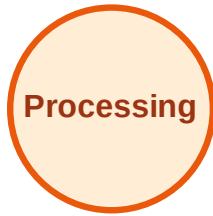
Visited: E



BFS Traversal

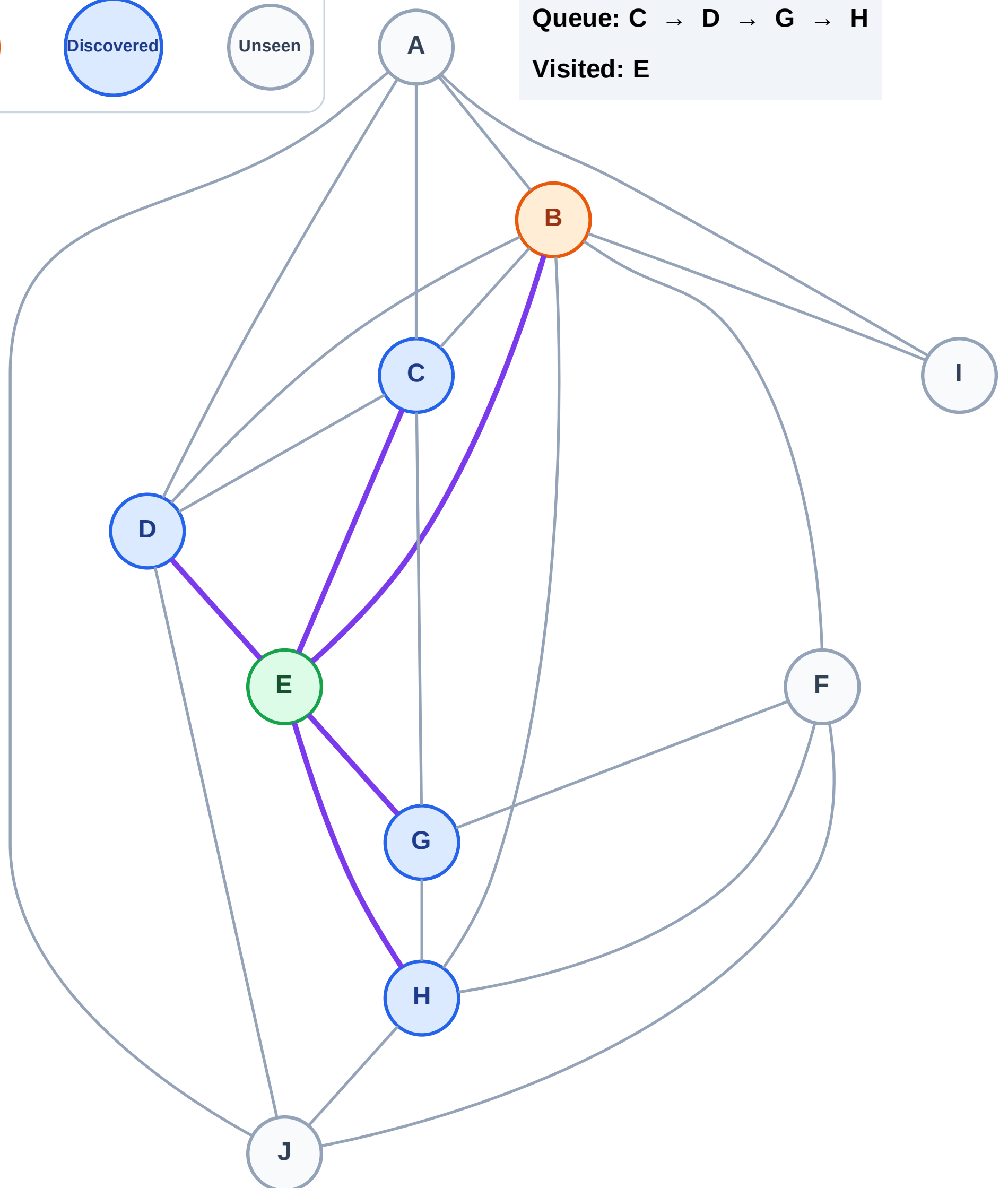
Step 4: Process node B

Legend



Queue: C → D → G → H

Visited: E



BFS Traversal

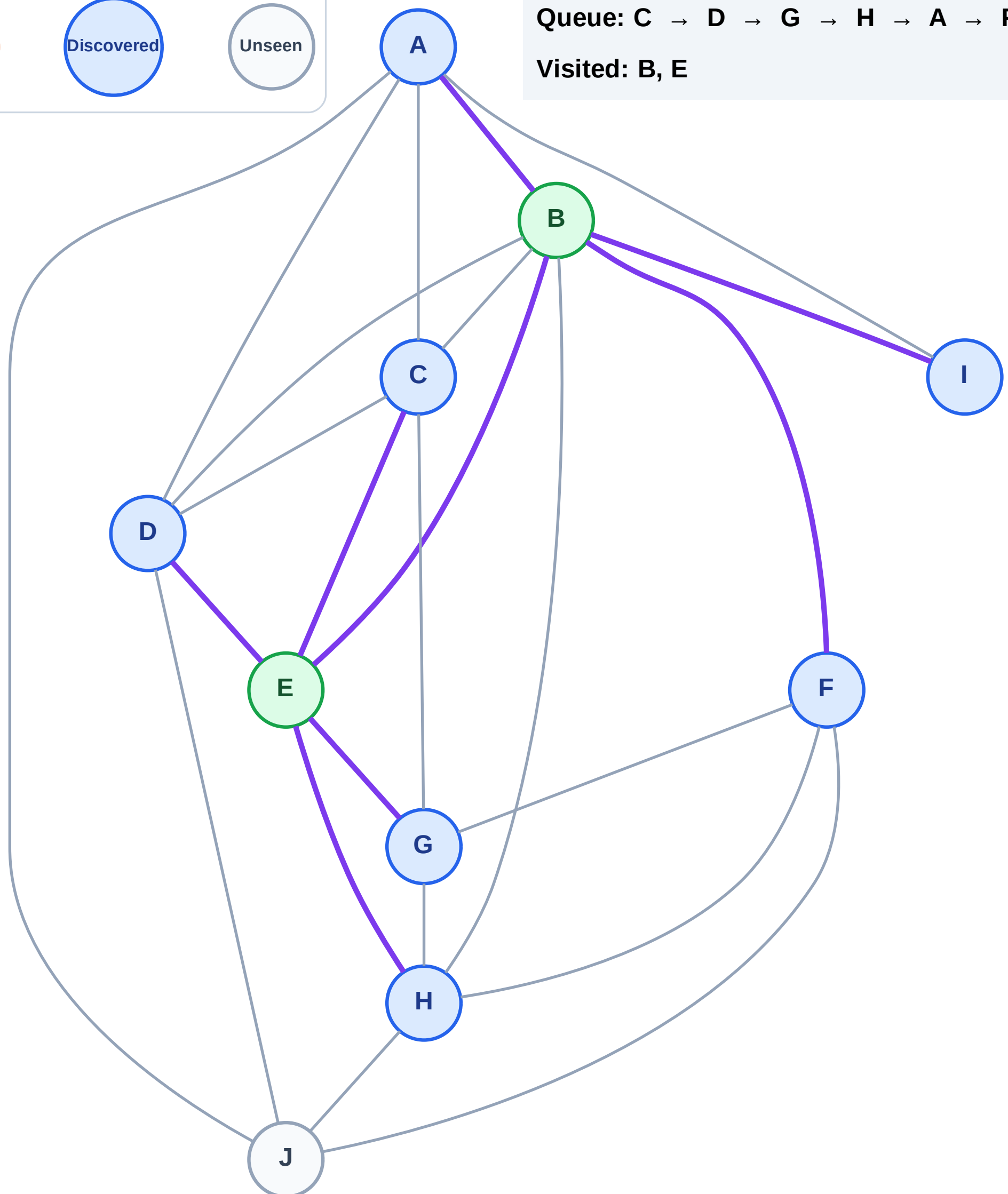
Step 5: Discover A, F, I from B

Legend



Queue: C → D → G → H → A → F → I

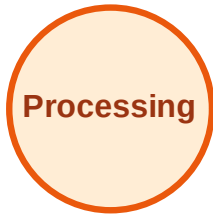
Visited: B, E



BFS Traversal

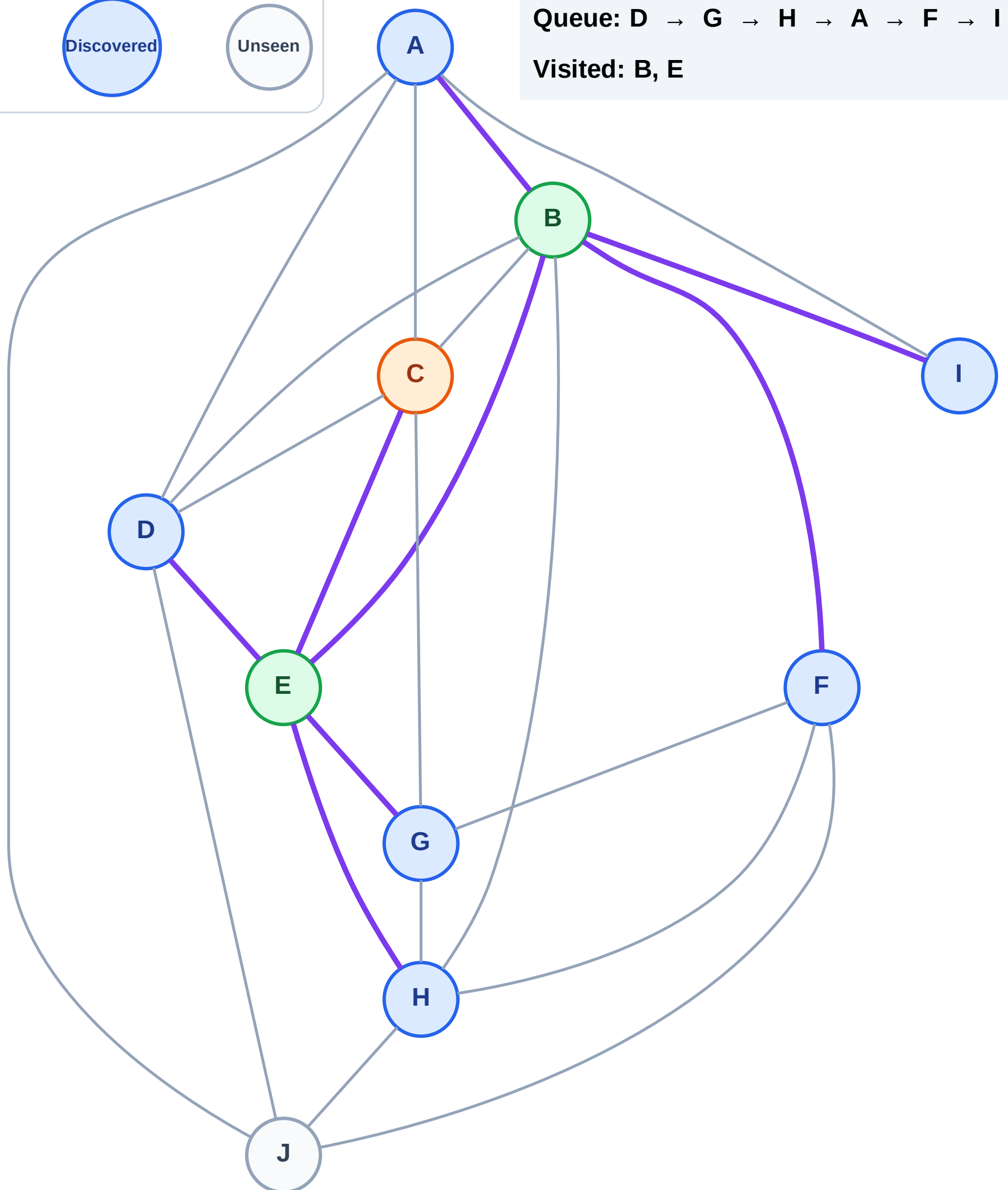
Step 6: Process node C

Legend



Queue: D → G → H → A → F → I

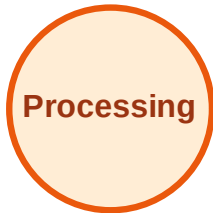
Visited: B, E



BFS Traversal

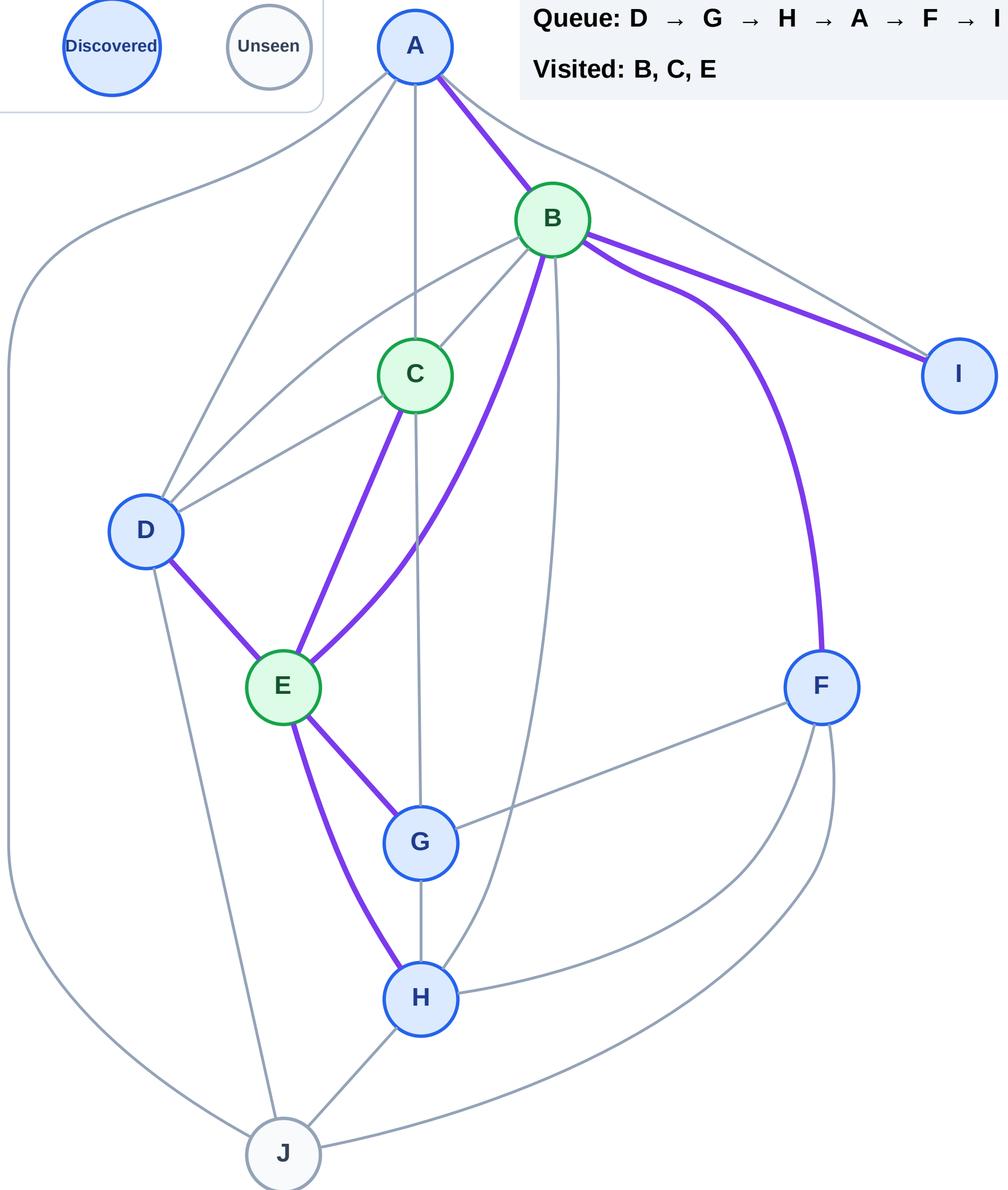
Step 7: No new neighbors from C

Legend



Queue: D → G → H → A → F → I

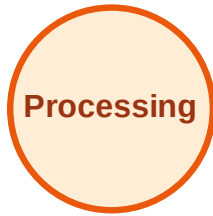
Visited: B, C, E



BFS Traversal

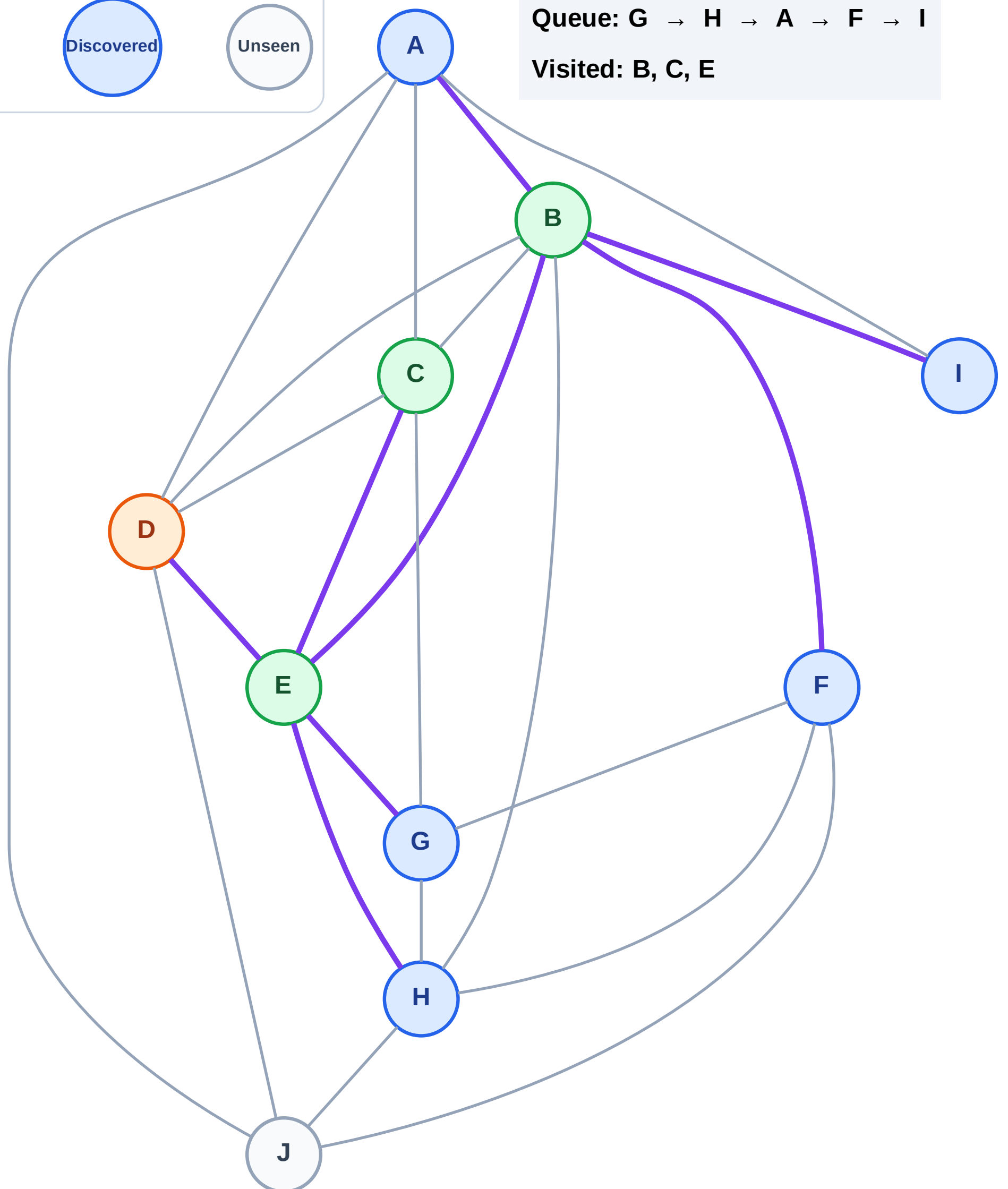
Step 8: Process node D

Legend



Queue: G → H → A → F → I

Visited: B, C, E



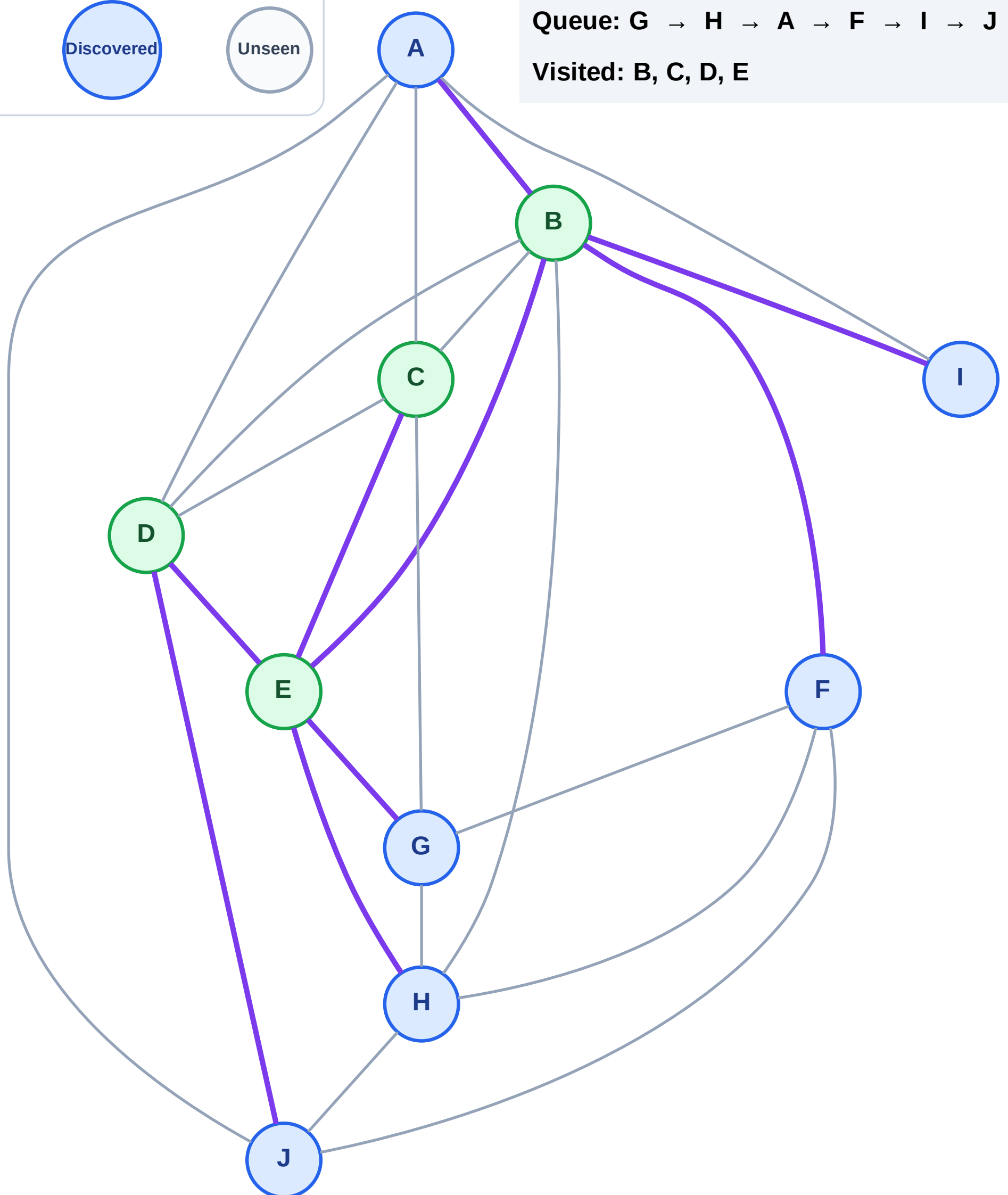
BFS Traversal

Step 9: Discover J from D

Legend



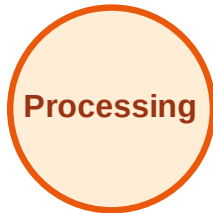
Queue: G → H → A → F → I → J
Visited: B, C, D, E



BFS Traversal

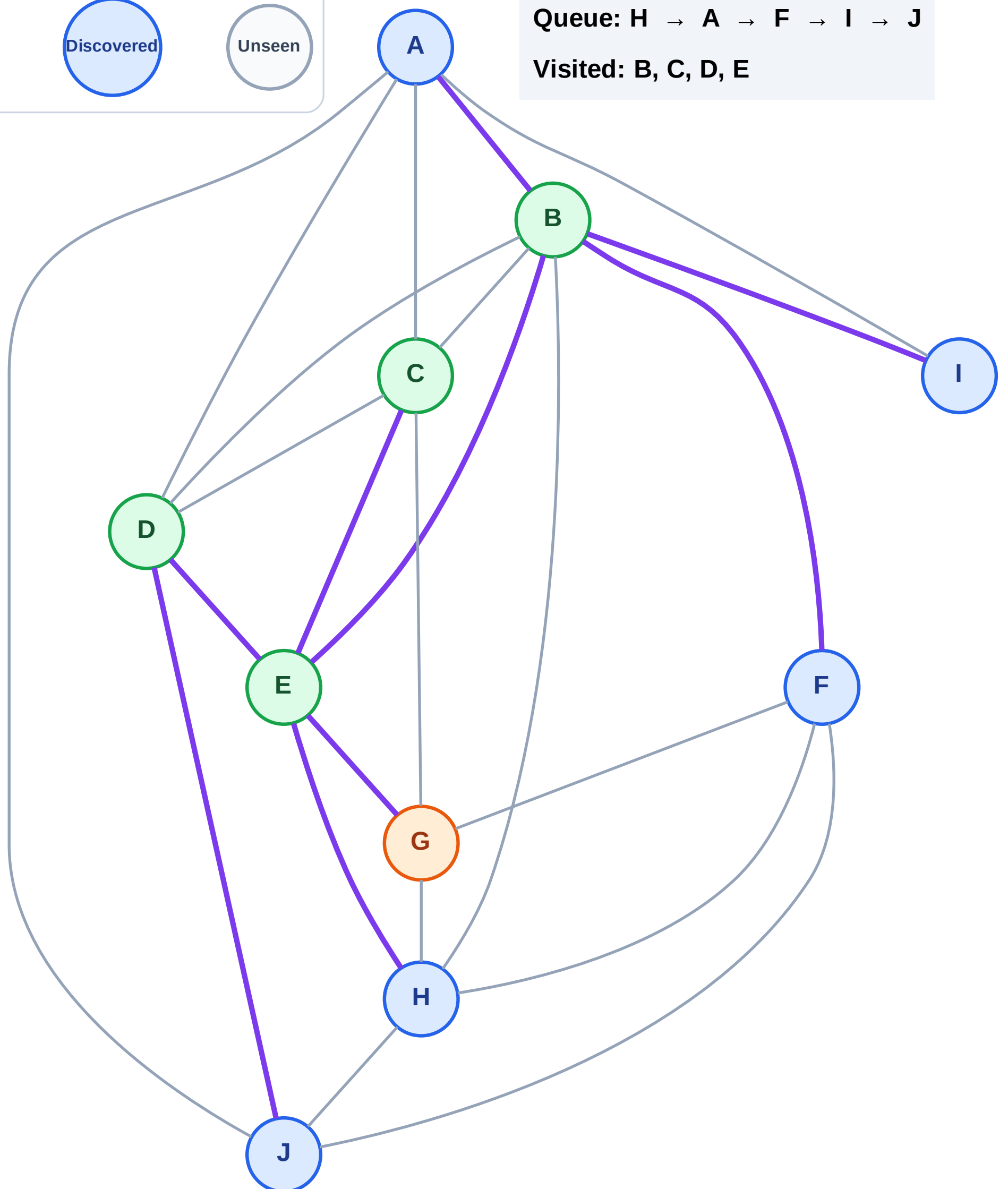
Step 10: Process node G

Legend



Queue: H → A → F → I → J

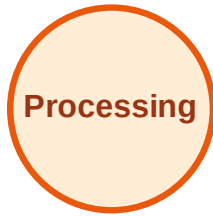
Visited: B, C, D, E



BFS Traversal

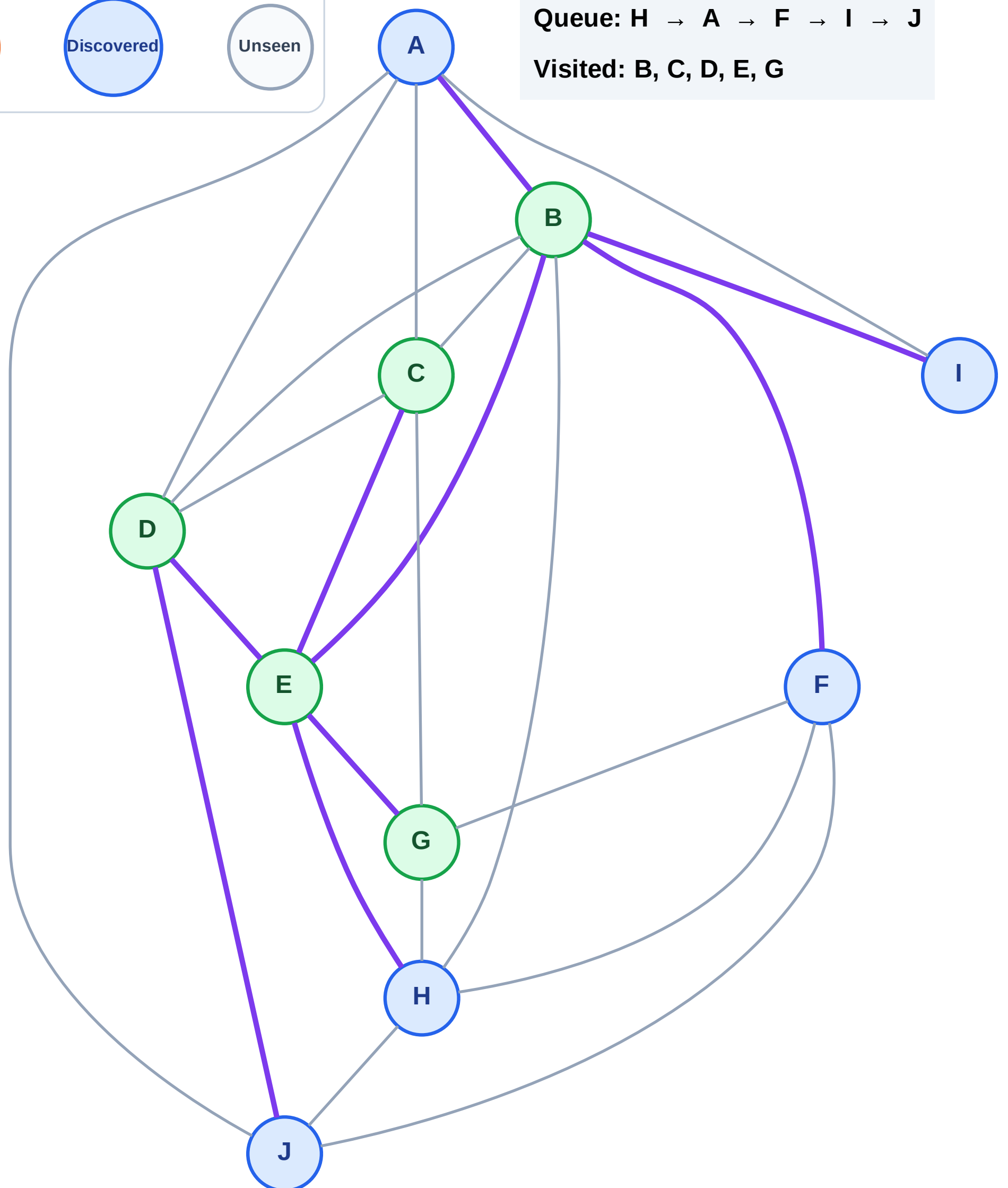
Step 11: No new neighbors from G

Legend



Queue: H → A → F → I → J

Visited: B, C, D, E, G



BFS Traversal

Step 12: Process node H

Legend

Complete

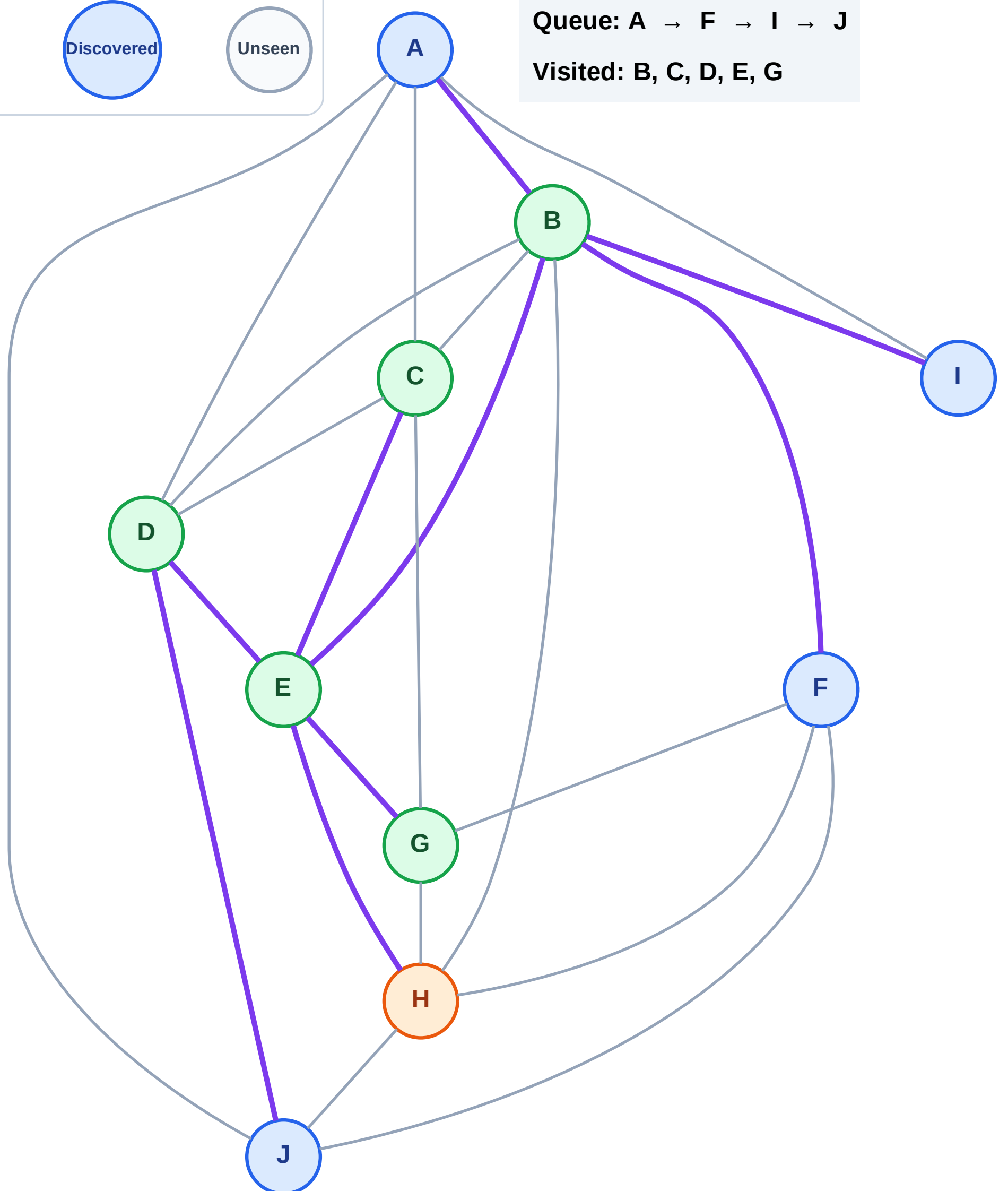
Processing

Discovered

Unseen

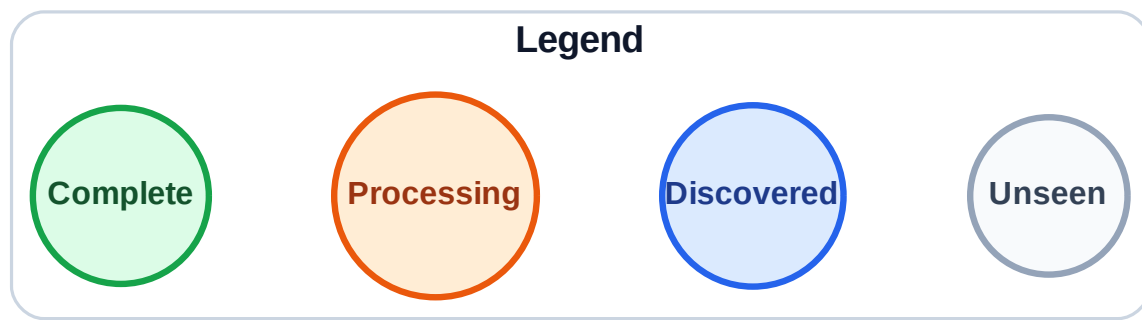
Queue: A → F → I → J

Visited: B, C, D, E, G



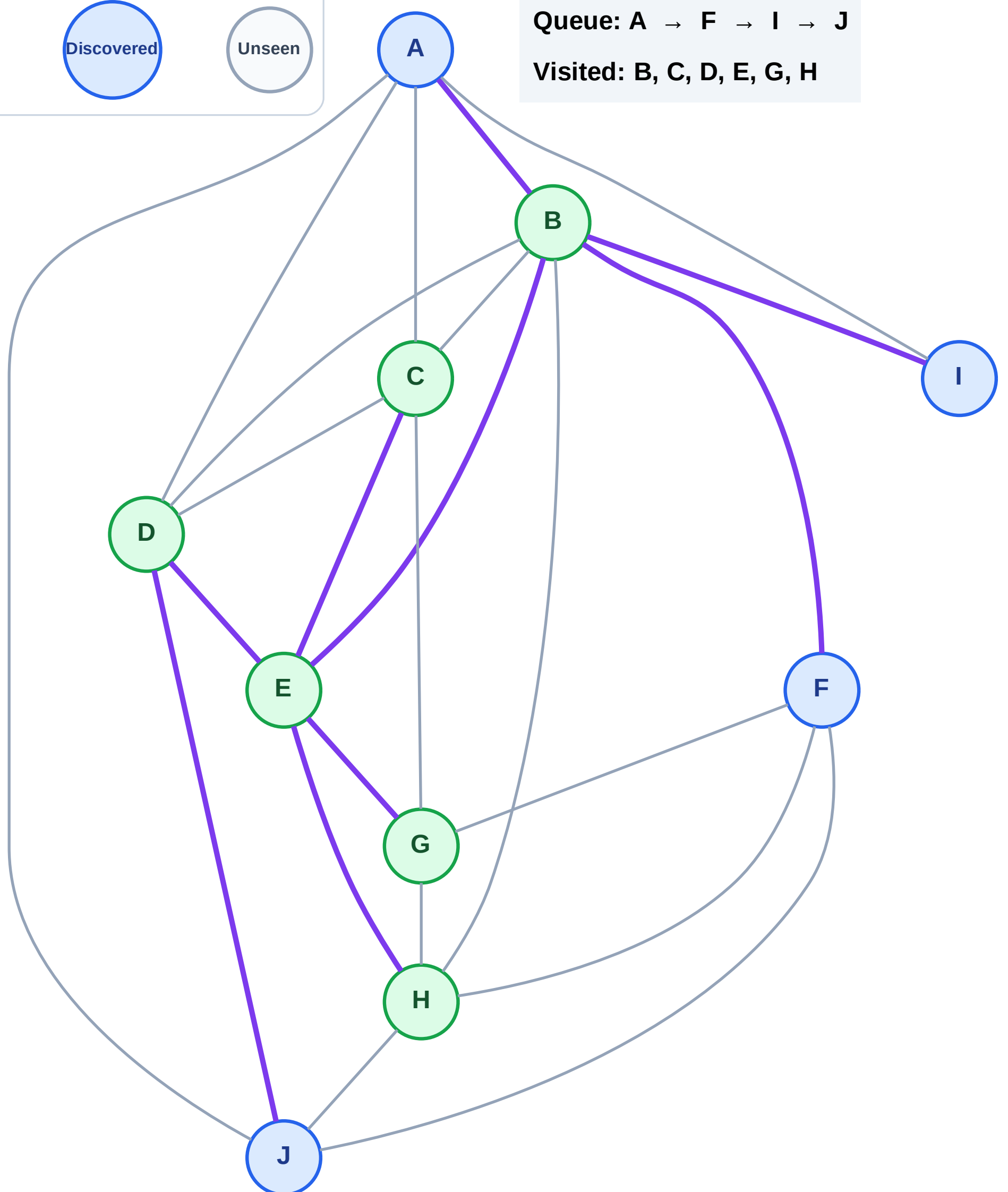
BFS Traversal

Step 13: No new neighbors from H



Queue: A → F → I → J

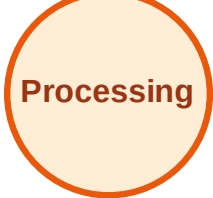
Visited: B, C, D, E, G, H



BFS Traversal

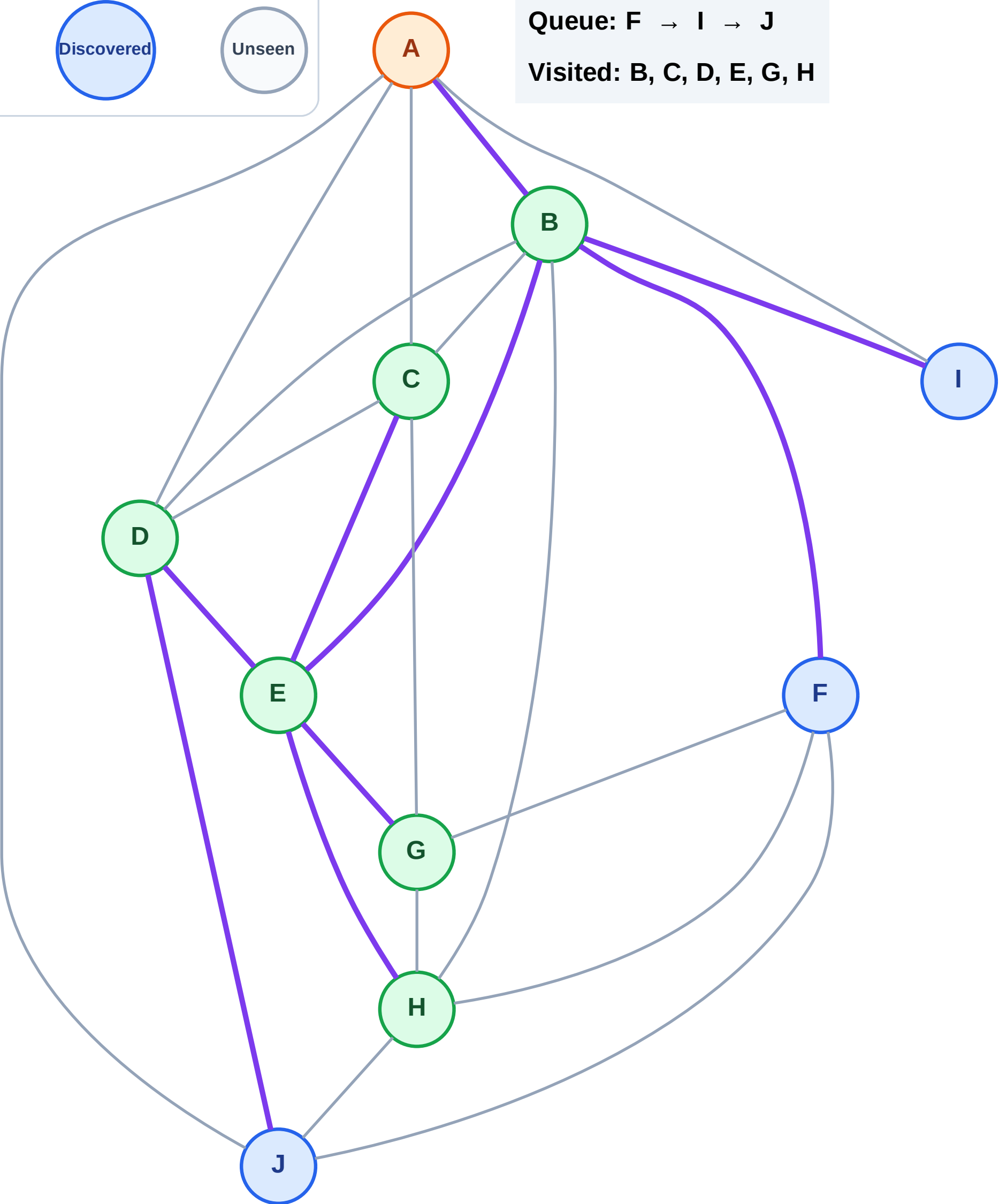
Step 14: Process node A

Legend



Queue: F → I → J

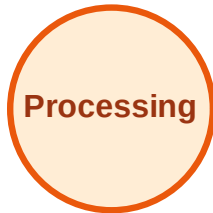
Visited: B, C, D, E, G, H



BFS Traversal

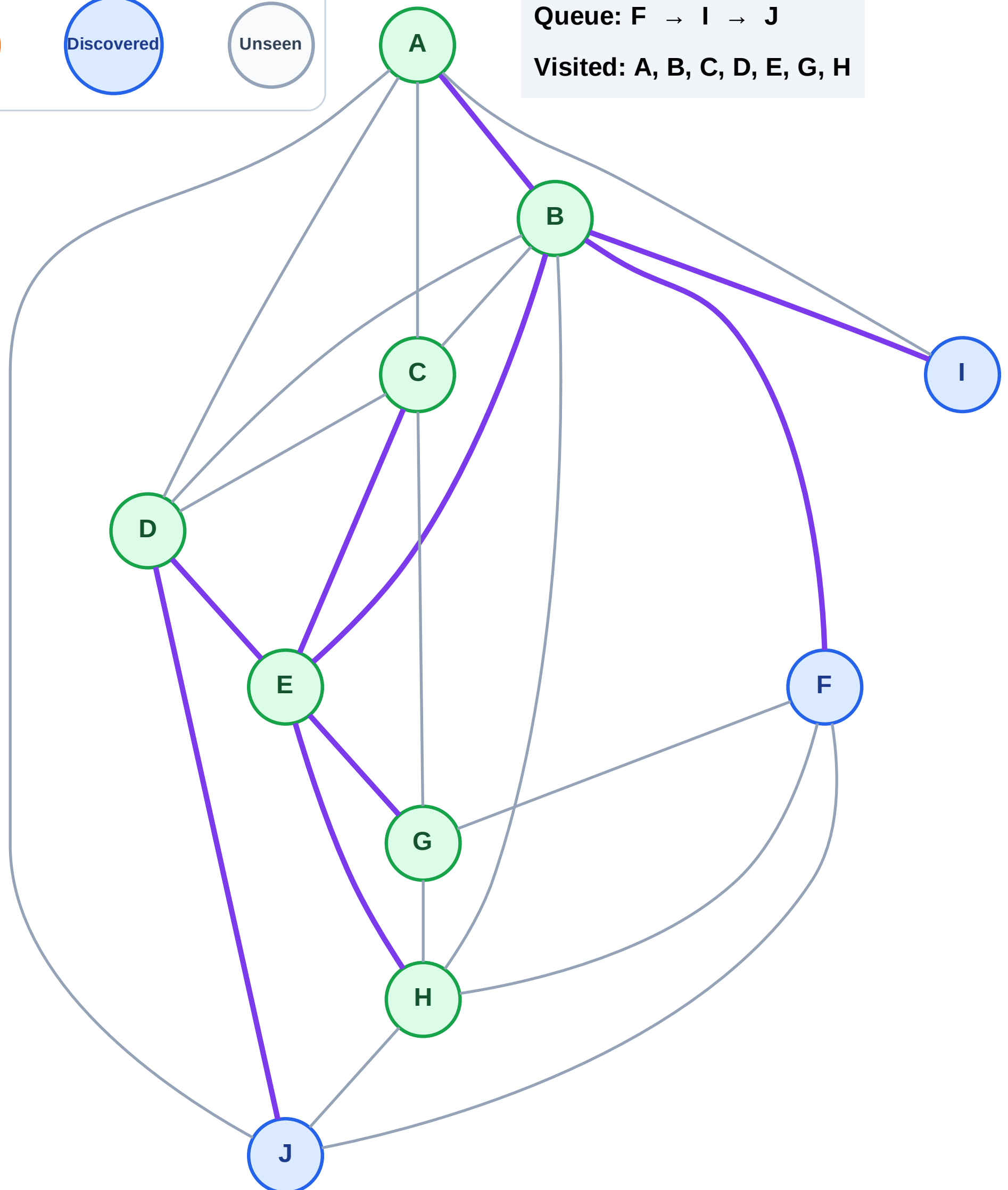
Step 15: No new neighbors from A

Legend



Queue: F → I → J

Visited: A, B, C, D, E, G, H



BFS Traversal

Step 16: Process node F

Legend

Complete

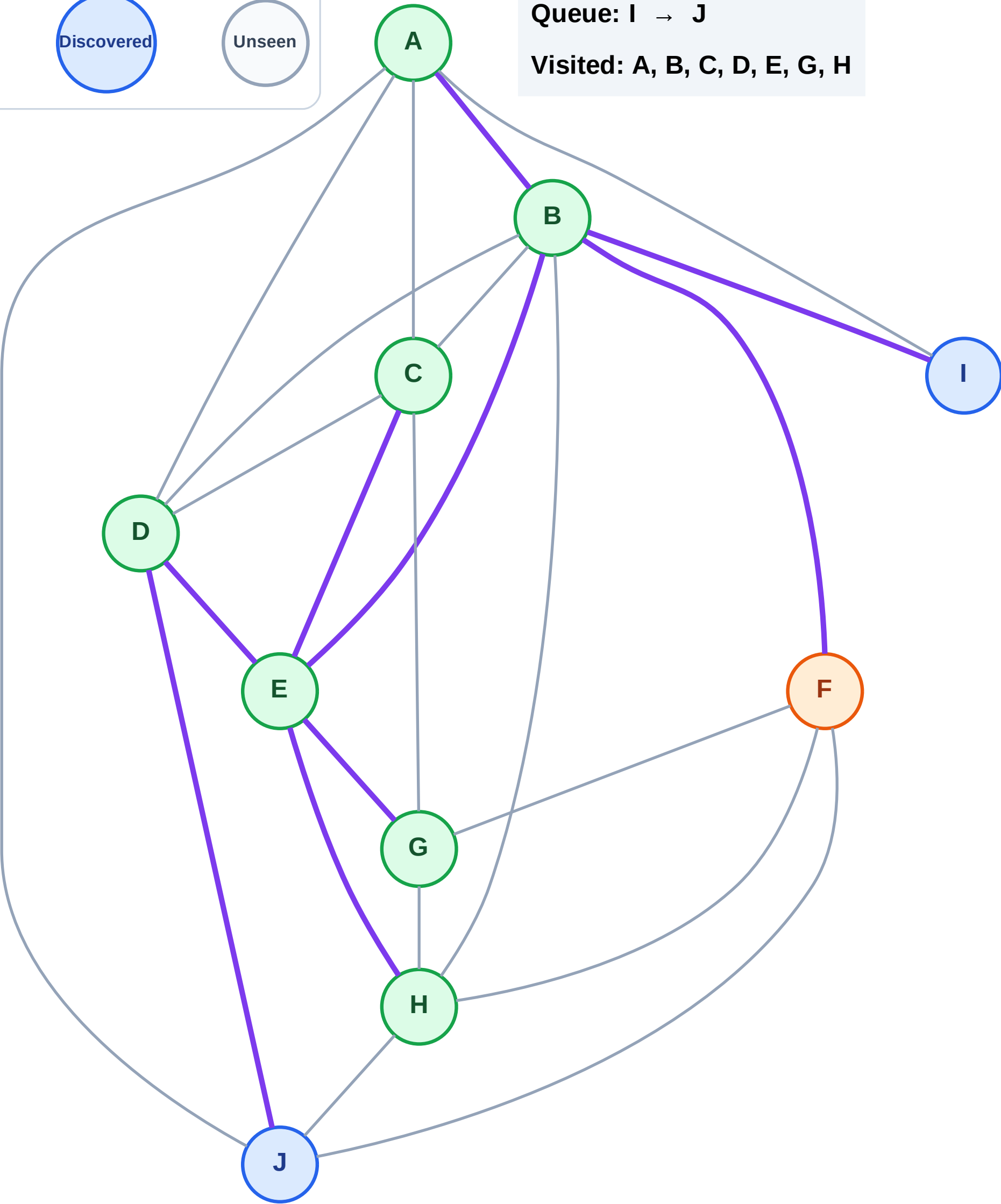
Processing

Discovered

Unseen

Queue: I → J

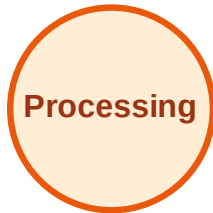
Visited: A, B, C, D, E, G, H



BFS Traversal

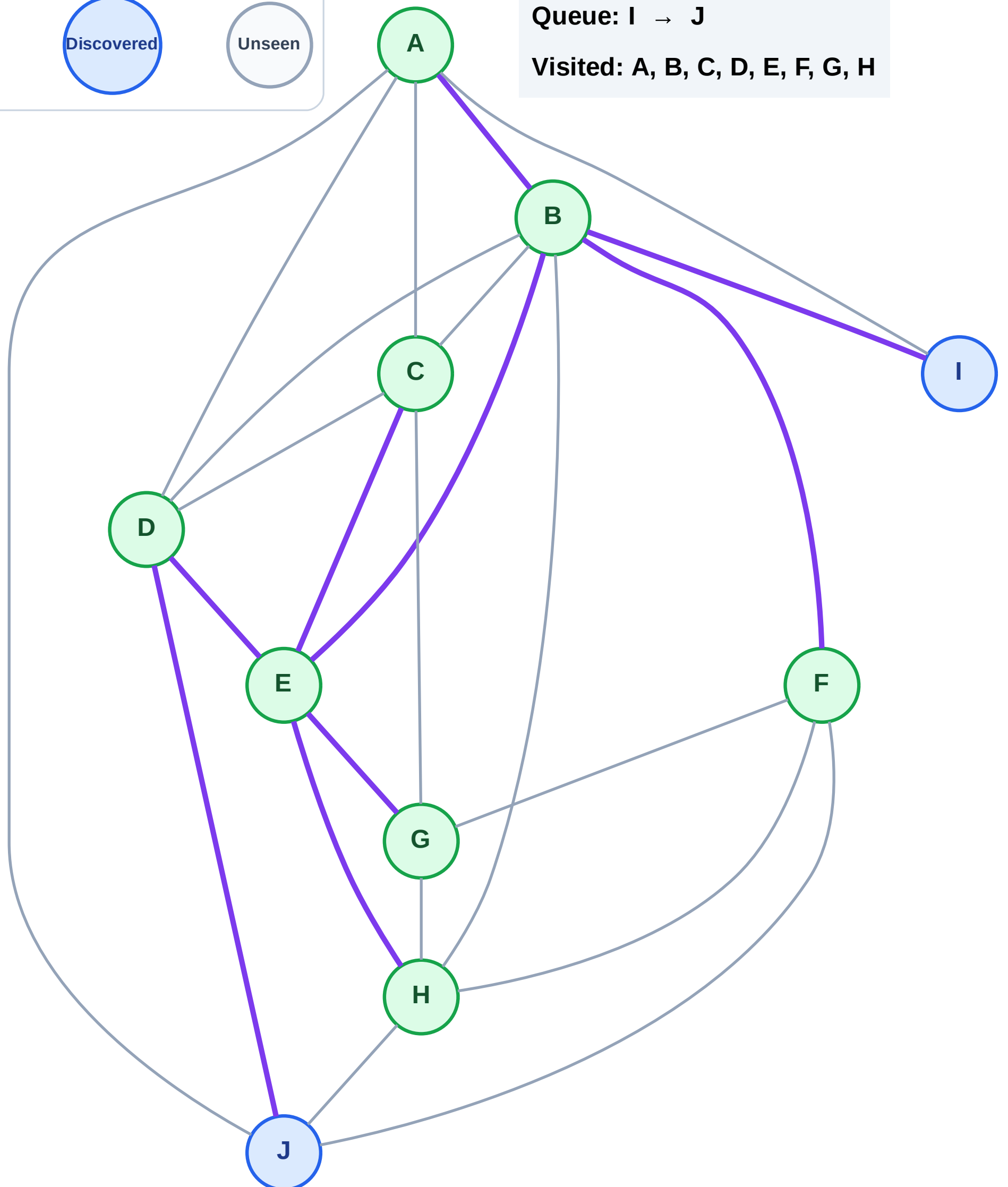
Step 17: No new neighbors from F

Legend



Queue: I → J

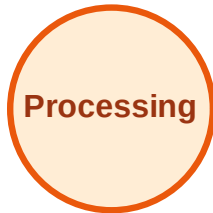
Visited: A, B, C, D, E, F, G, H



BFS Traversal

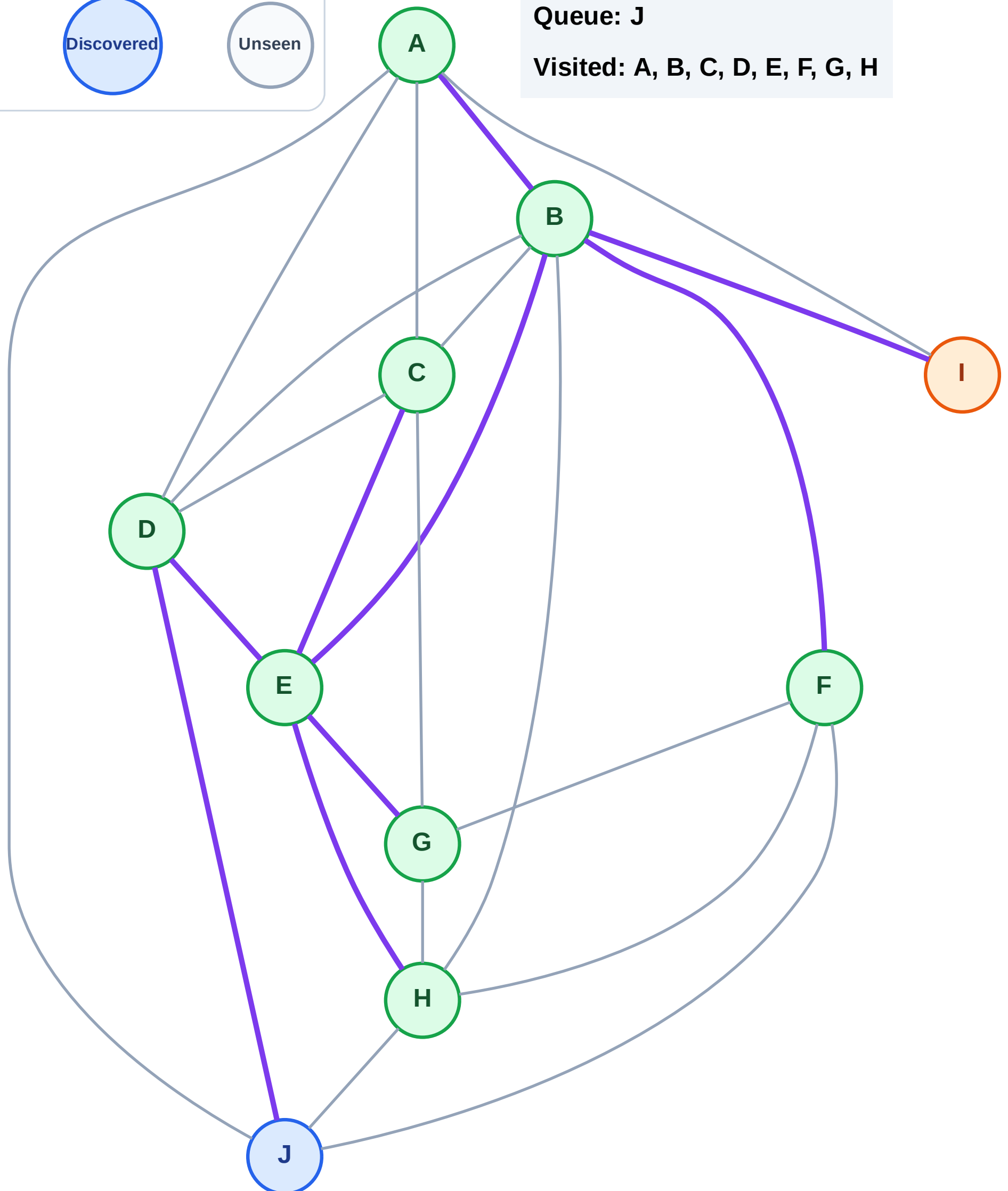
Step 18: Process node I

Legend



Queue: J

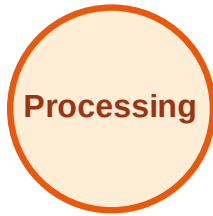
Visited: A, B, C, D, E, F, G, H



BFS Traversal

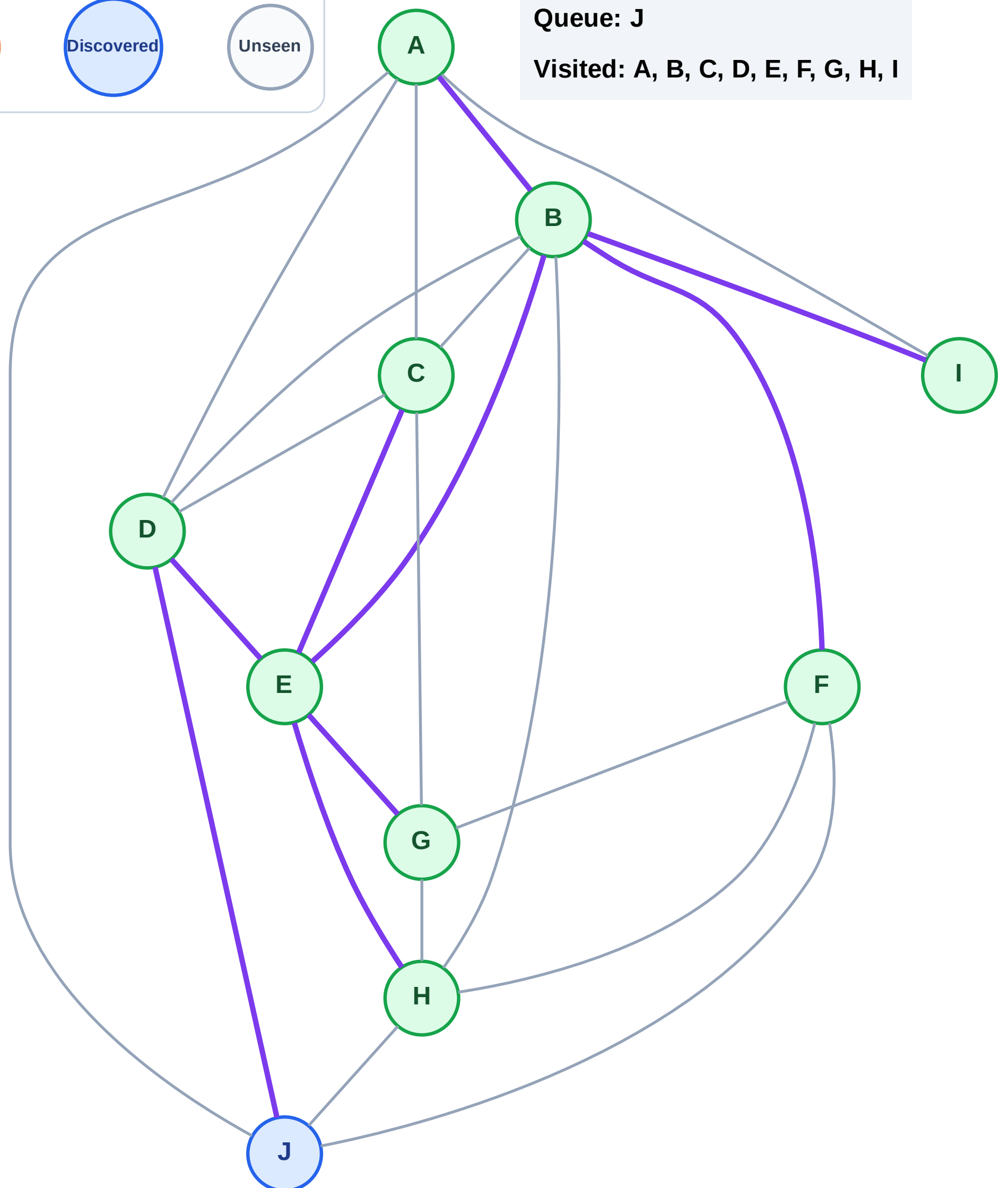
Step 19: No new neighbors from I

Legend



Queue: J

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

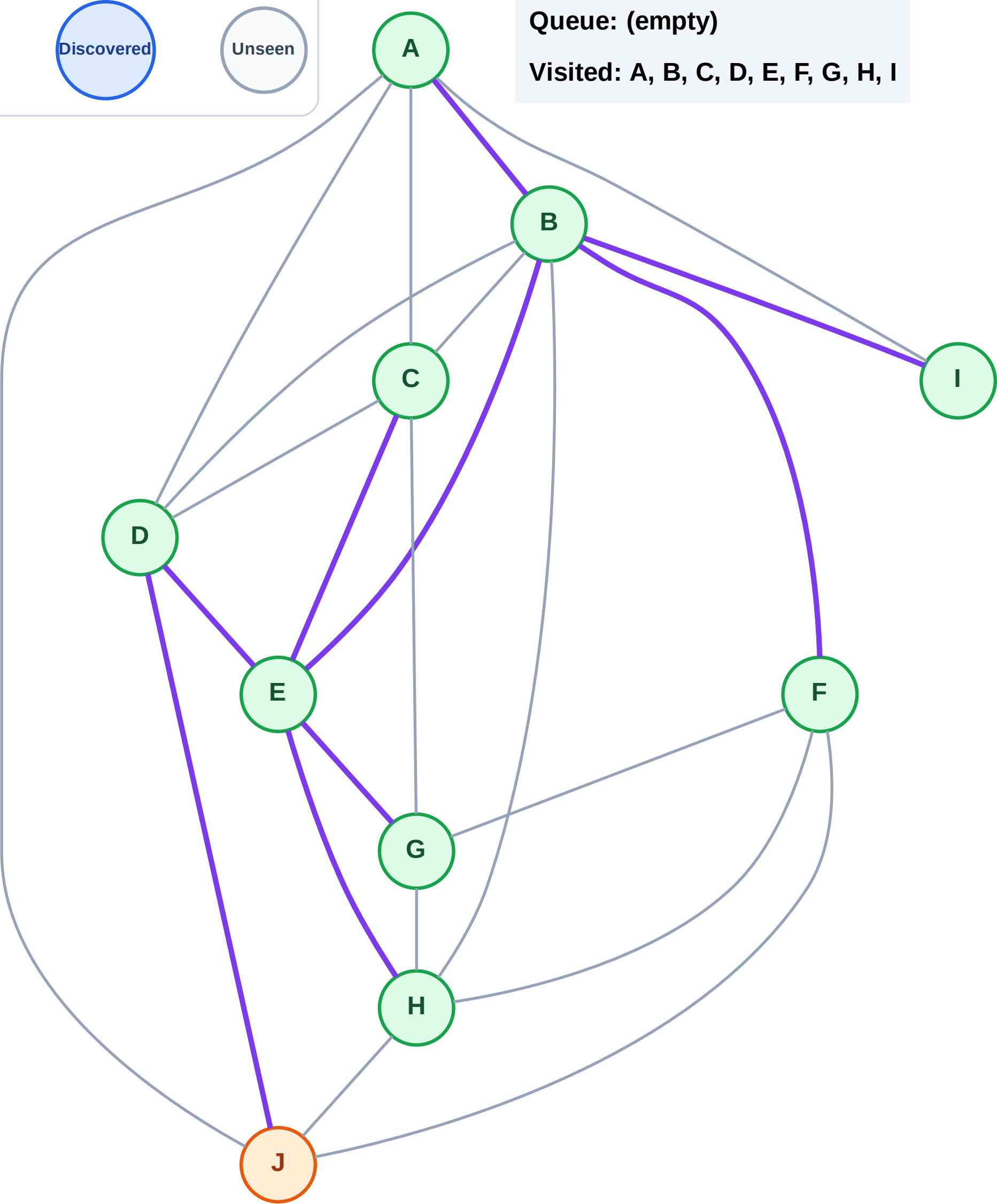
Step 20: Process node J

Legend



Queue: (empty)

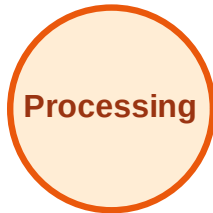
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

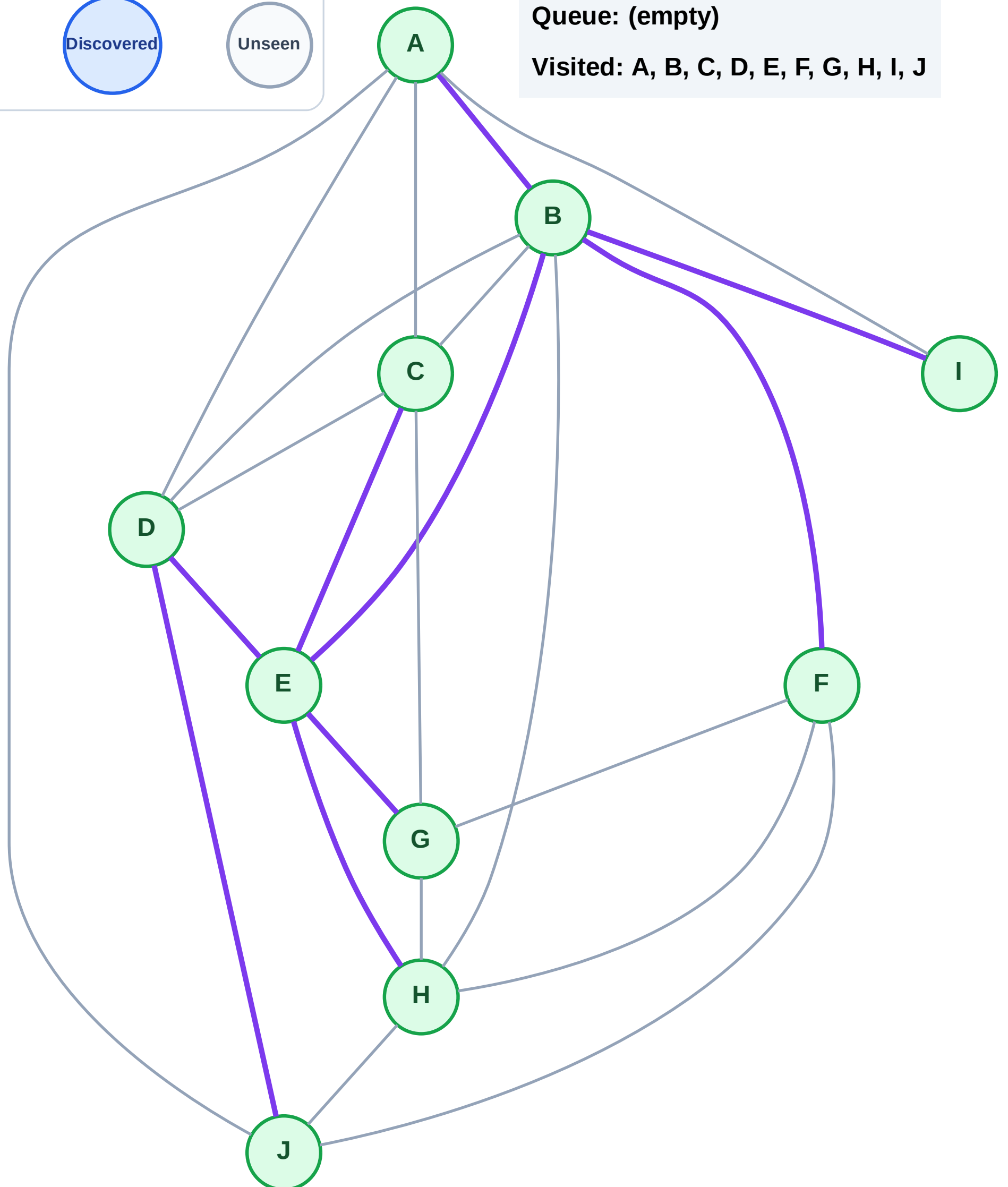
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at E

Legend

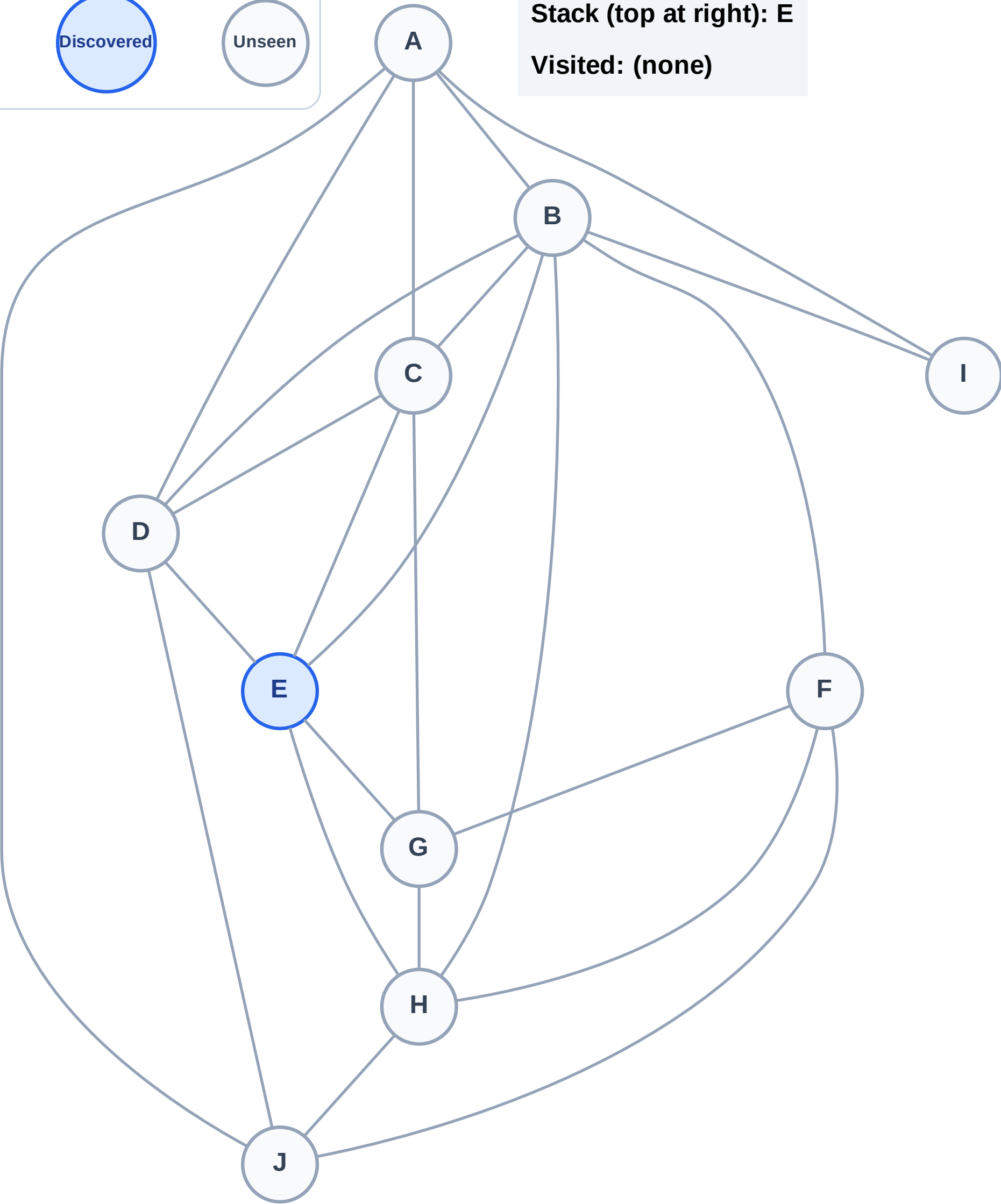
Complete

Processing

Discovered

Unseen

Stack (top at right): E
Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

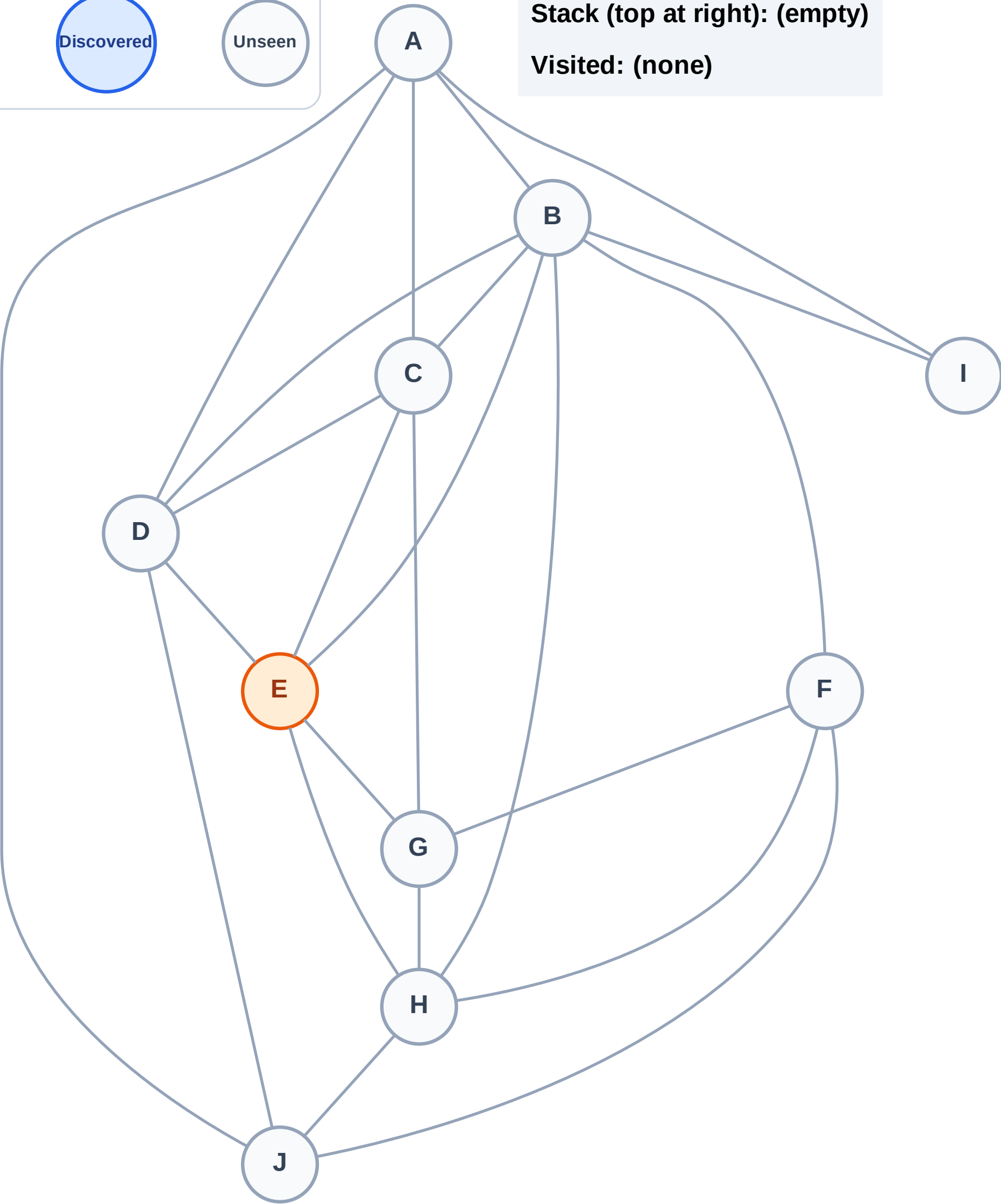
Complete

Processing

Discovered

Unseen

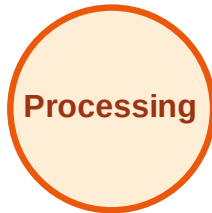
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

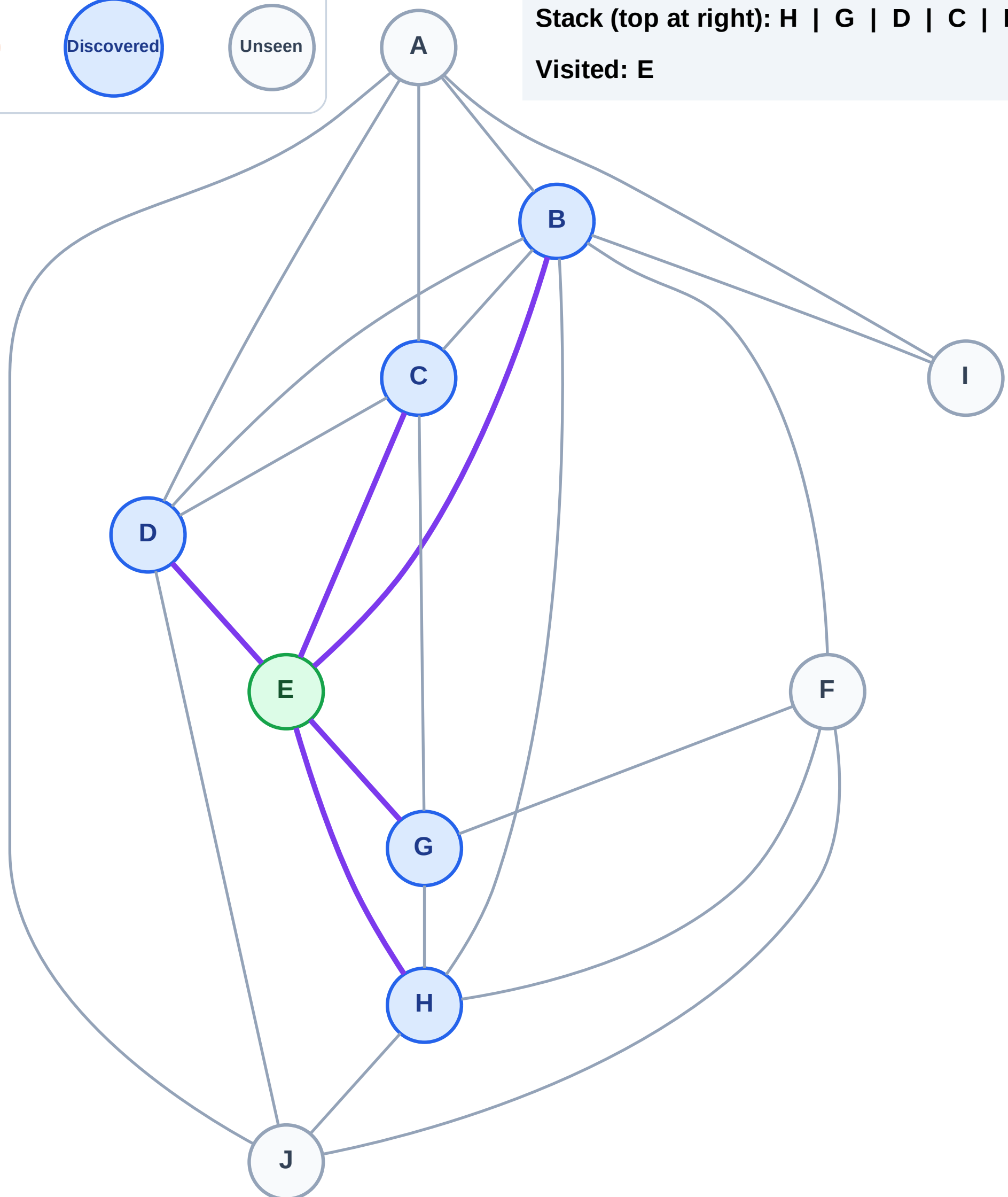
Step 3: Discover H, G, D, C, B from E

Legend



Stack (top at right): H | G | D | C | B

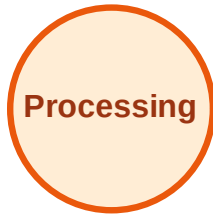
Visited: E



DFS Traversal

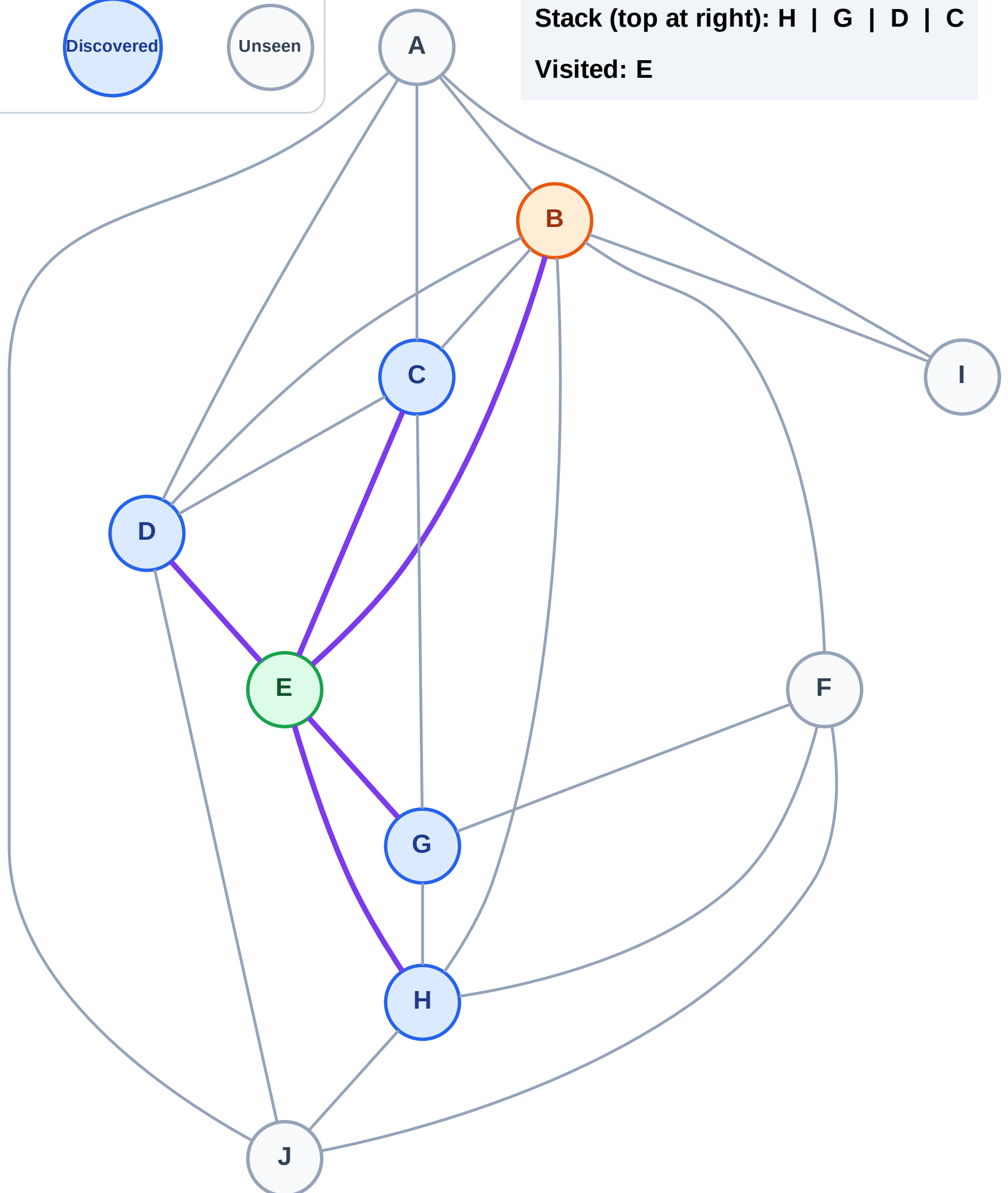
Step 4: Process node B

Legend



Stack (top at right): H | G | D | C

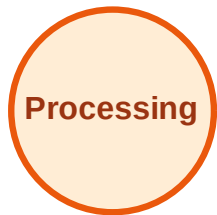
Visited: E



DFS Traversal

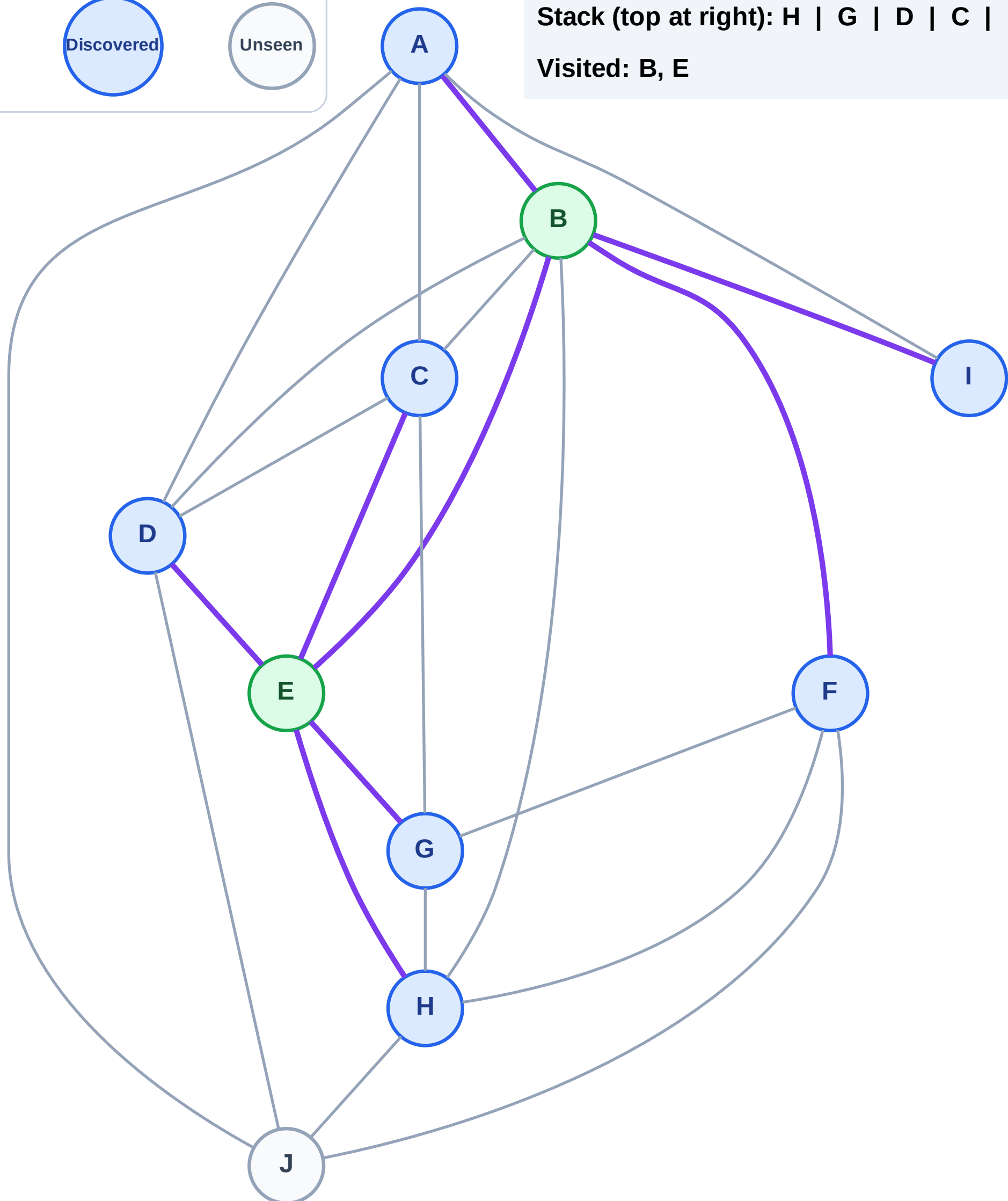
Step 5: Discover I, F, A from B

Legend



Stack (top at right): H | G | D | C | I | F | A

Visited: B, E



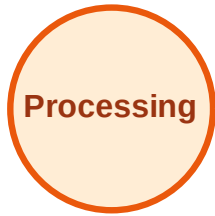
DFS Traversal

Step 6: Process node A

Legend



Complete



Processing



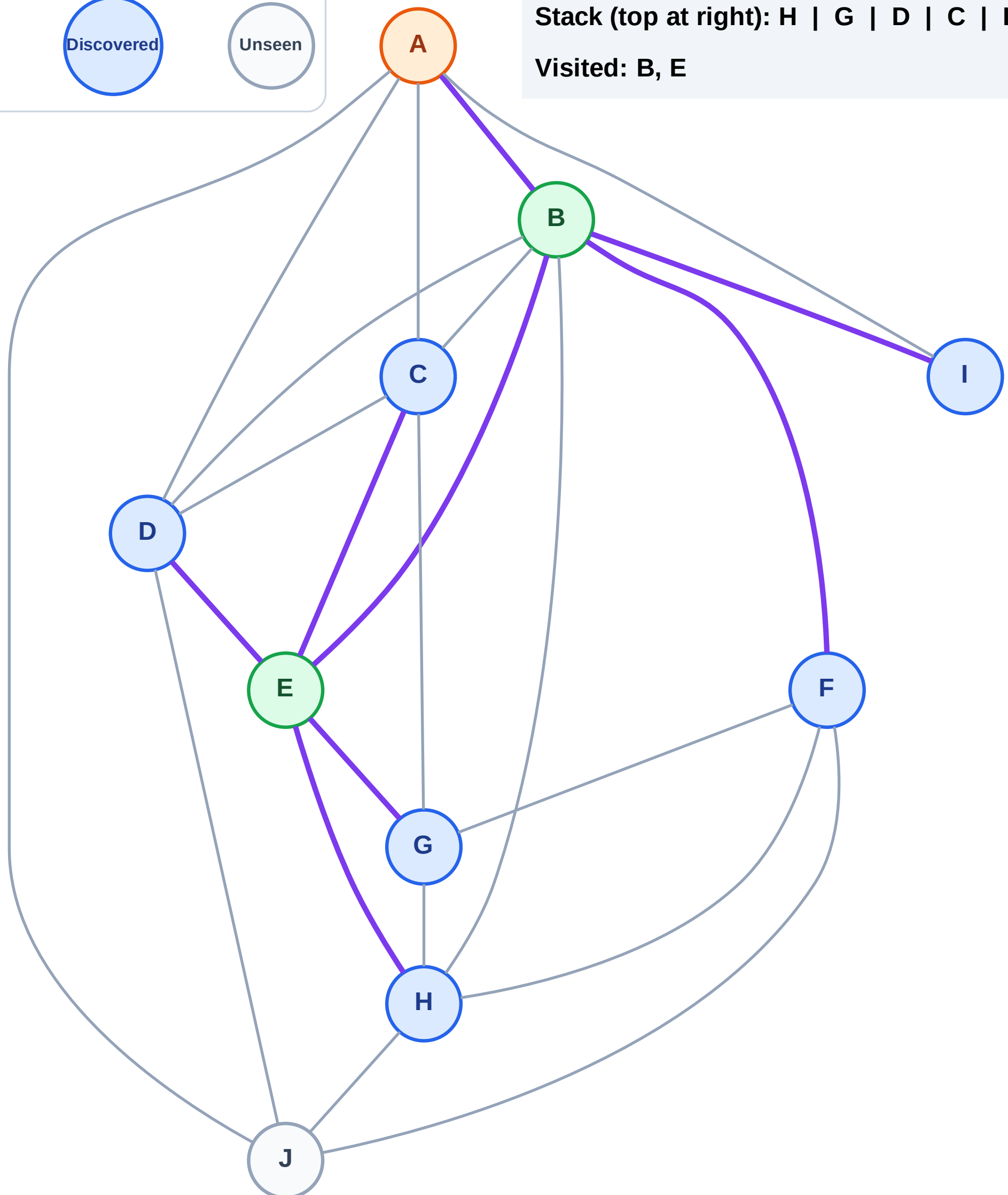
Discovered



Unseen

Stack (top at right): H | G | D | C | I | F

Visited: B, E



DFS Traversal

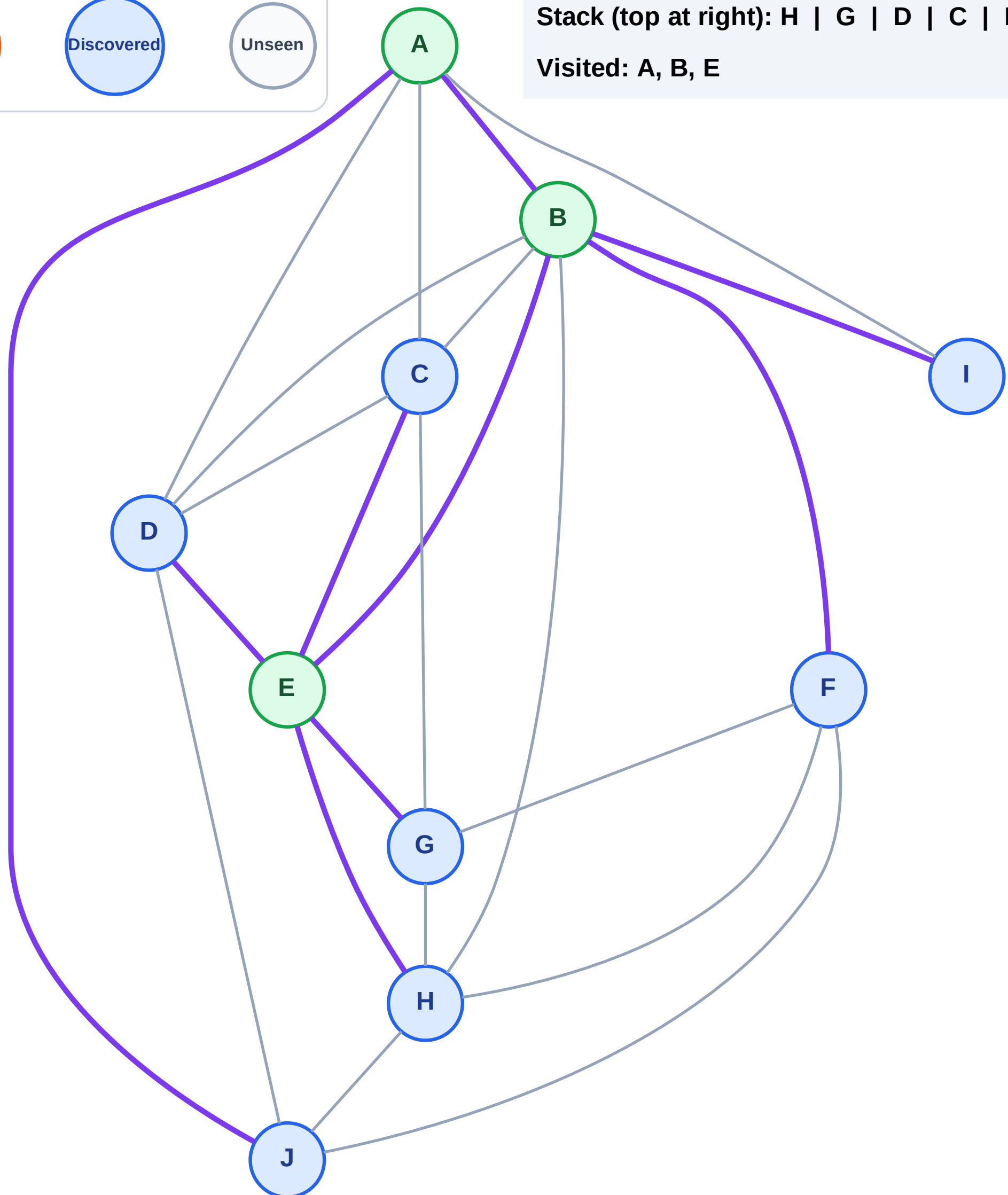
Step 7: Discover J from A

Legend



Stack (top at right): H | G | D | C | I | F | J

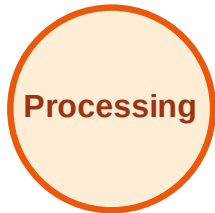
Visited: A, B, E



DFS Traversal

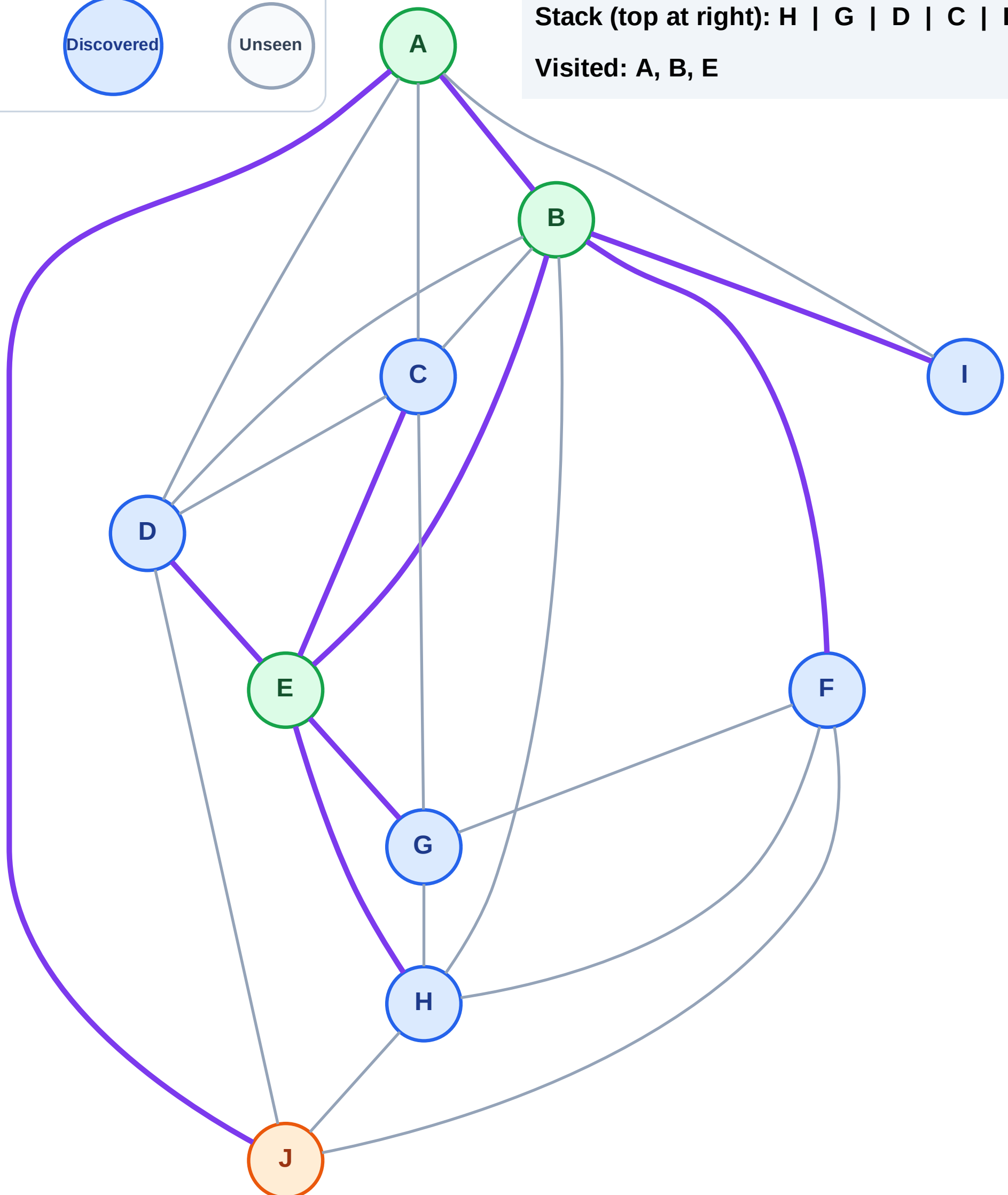
Step 8: Process node J

Legend



Stack (top at right): H | G | D | C | I | F

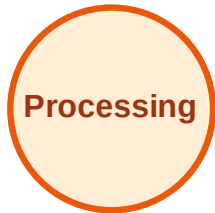
Visited: A, B, E



DFS Traversal

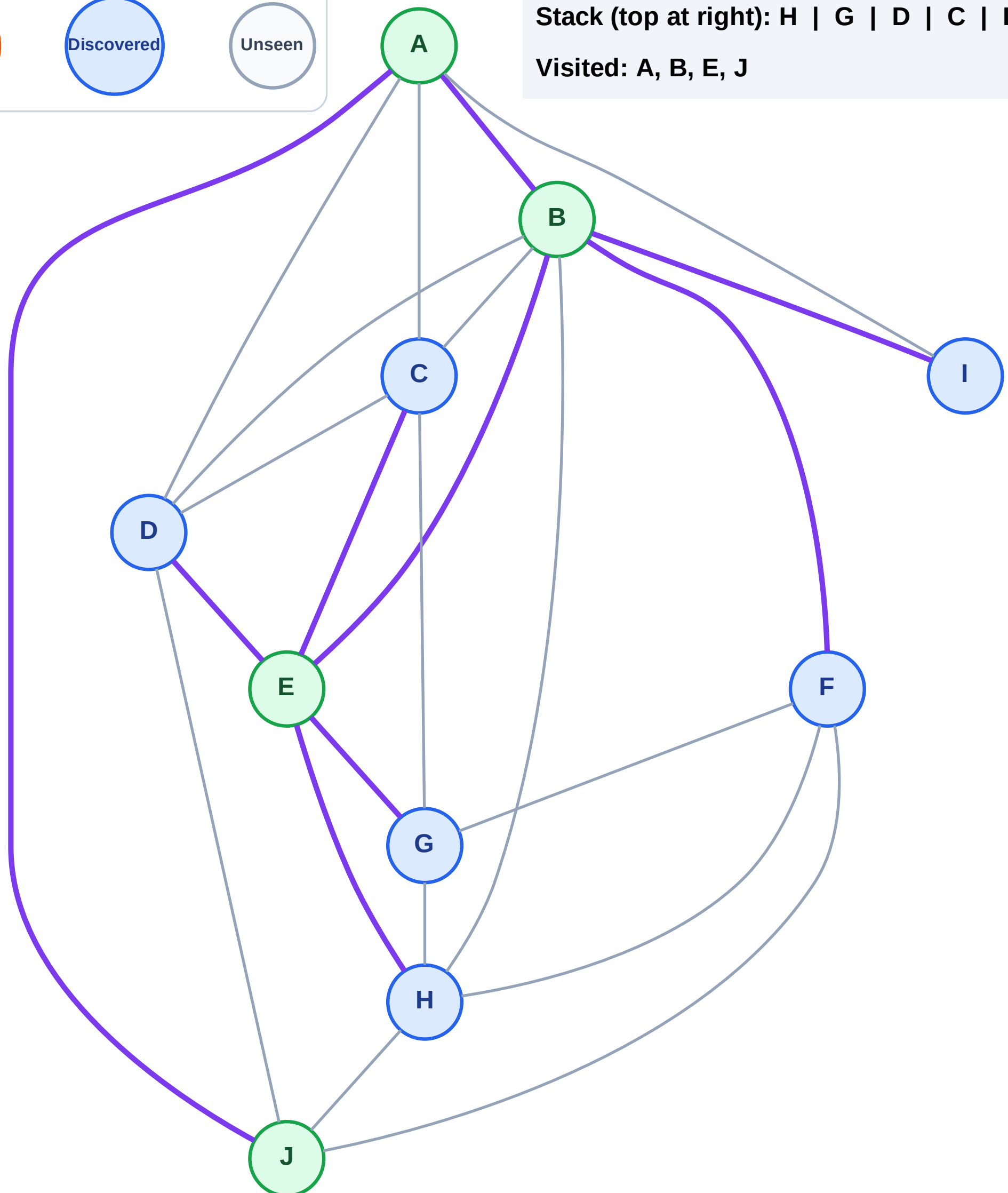
Step 9: No new neighbors from J

Legend



Stack (top at right): H | G | D | C | I | F

Visited: A, B, E, J



DFS Traversal

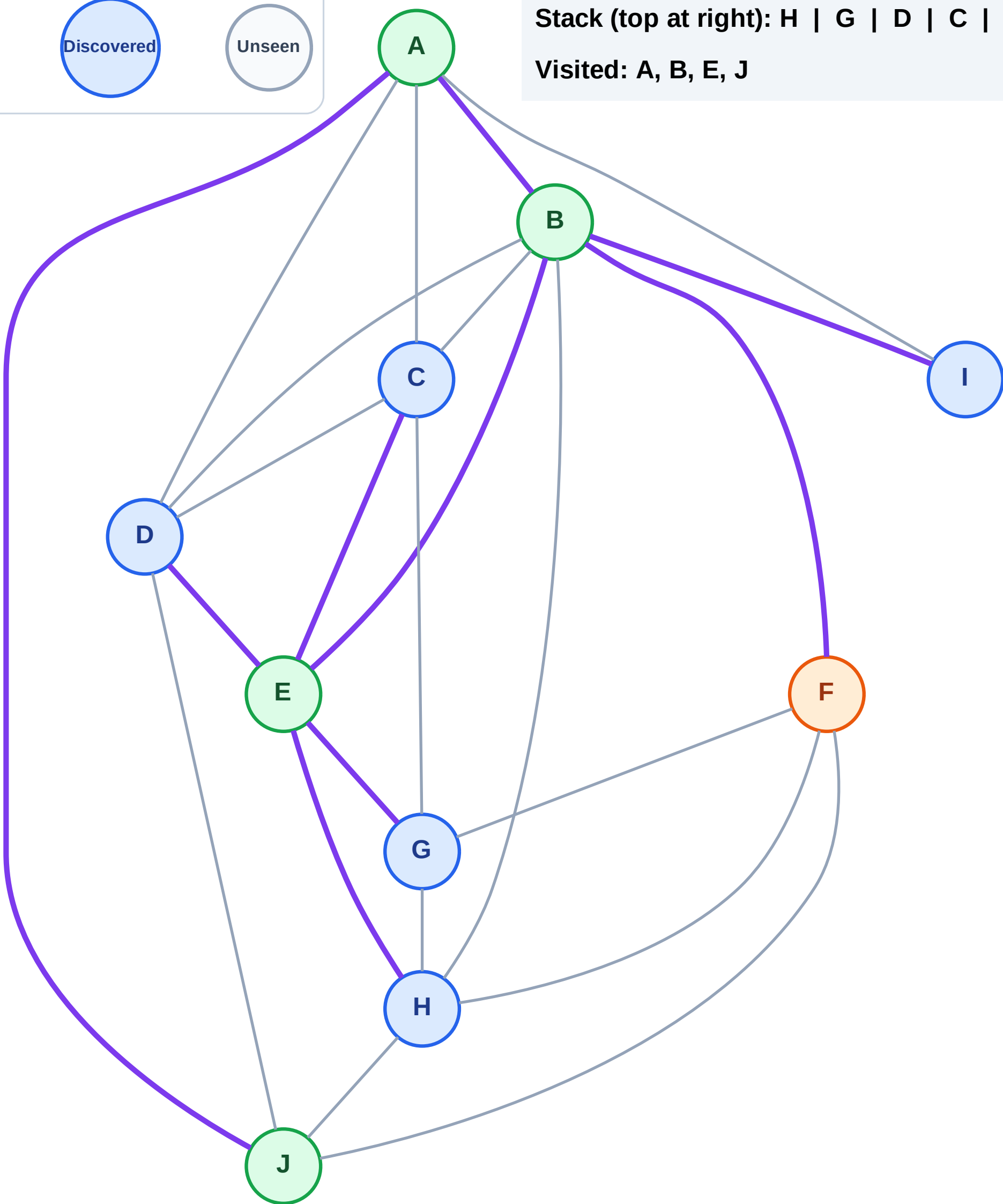
Step 10: Process node F

Legend



Stack (top at right): H | G | D | C | I

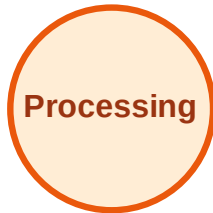
Visited: A, B, E, J



DFS Traversal

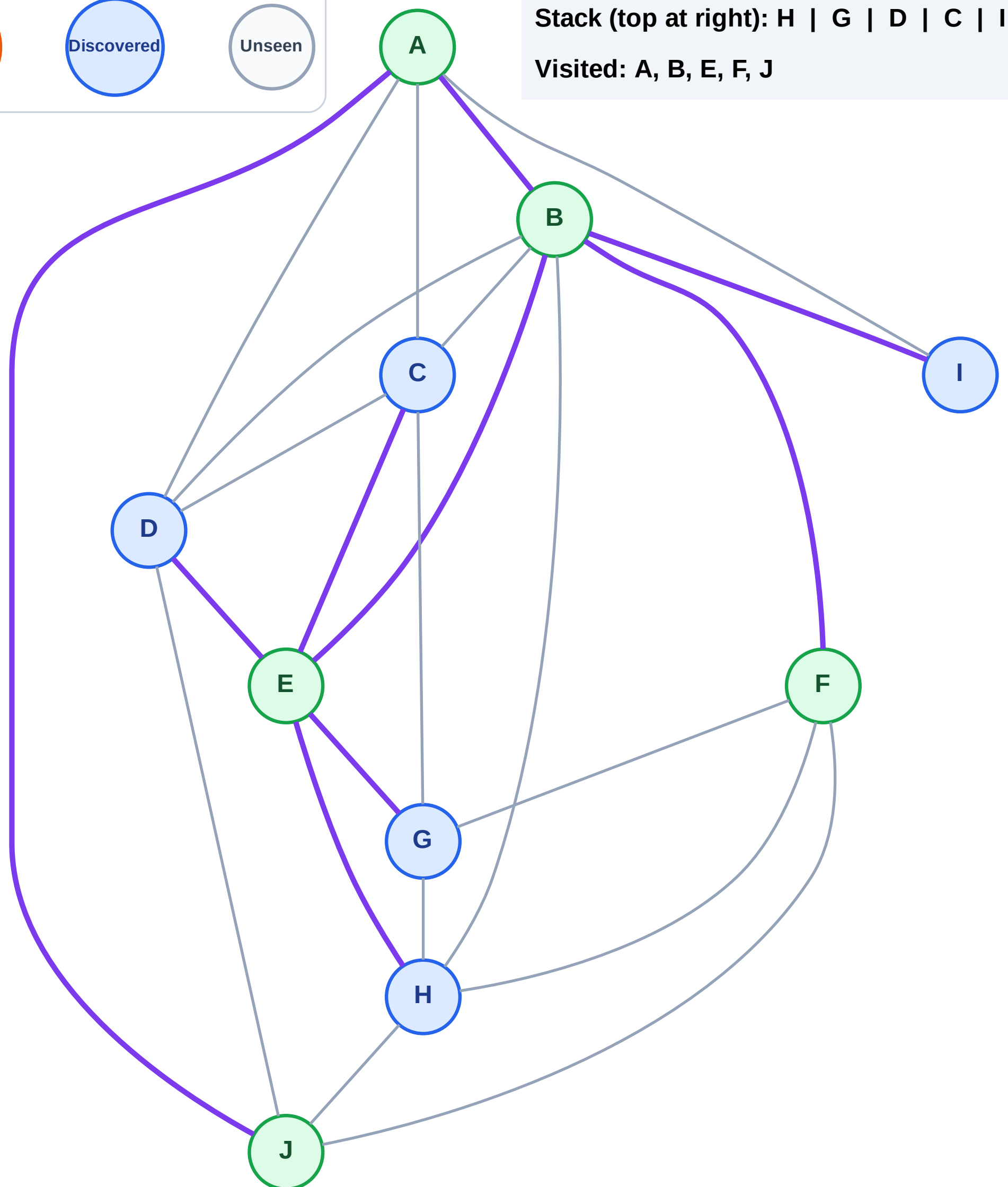
Step 11: No new neighbors from F

Legend



Stack (top at right): H | G | D | C | I

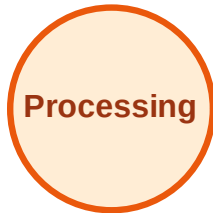
Visited: A, B, E, F, J



DFS Traversal

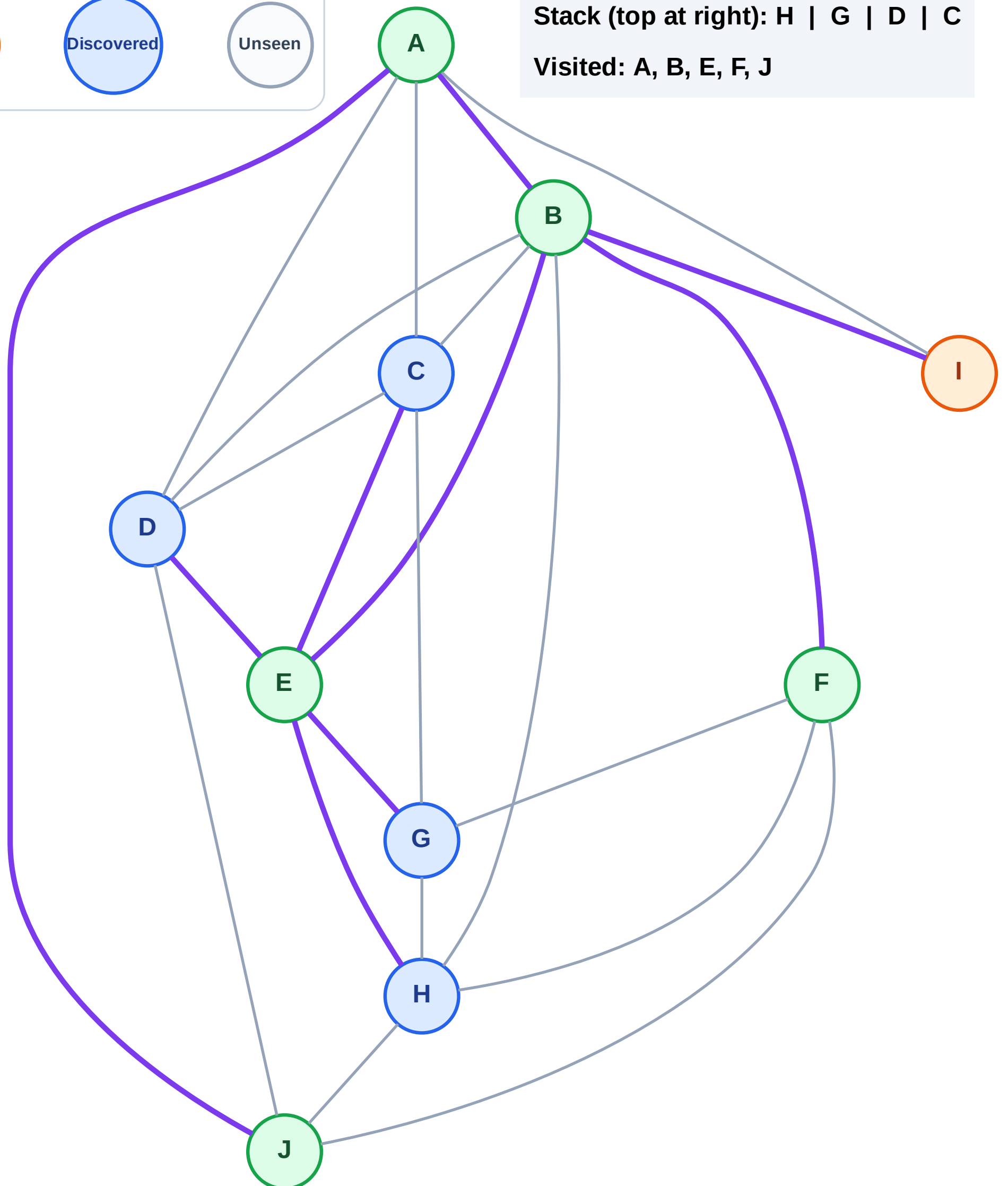
Step 12: Process node I

Legend



Stack (top at right): H | G | D | C

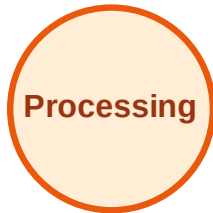
Visited: A, B, E, F, J



DFS Traversal

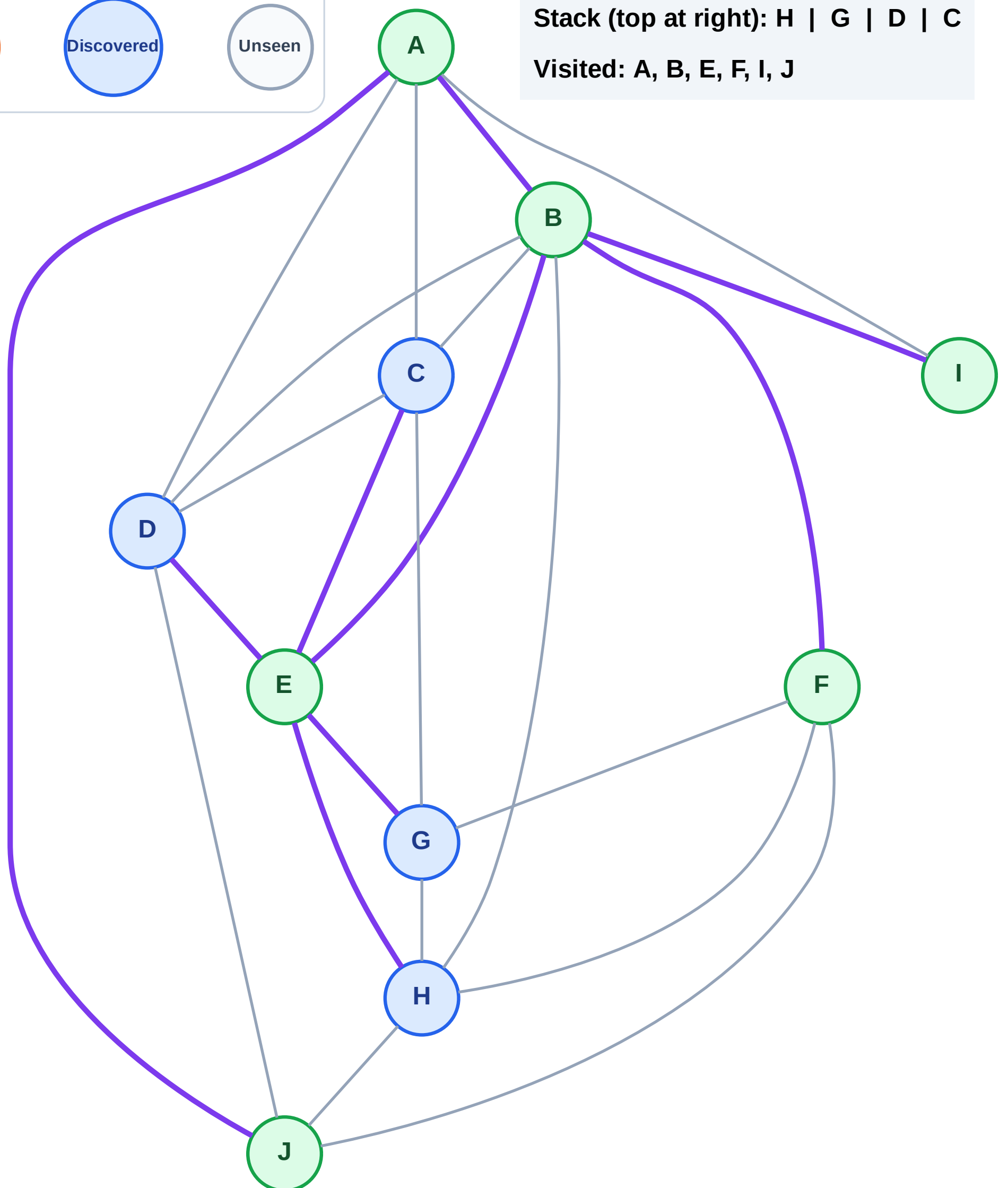
Step 13: No new neighbors from I

Legend



Stack (top at right): H | G | D | C

Visited: A, B, E, F, I, J



DFS Traversal

Step 14: Process node C

Legend

Complete

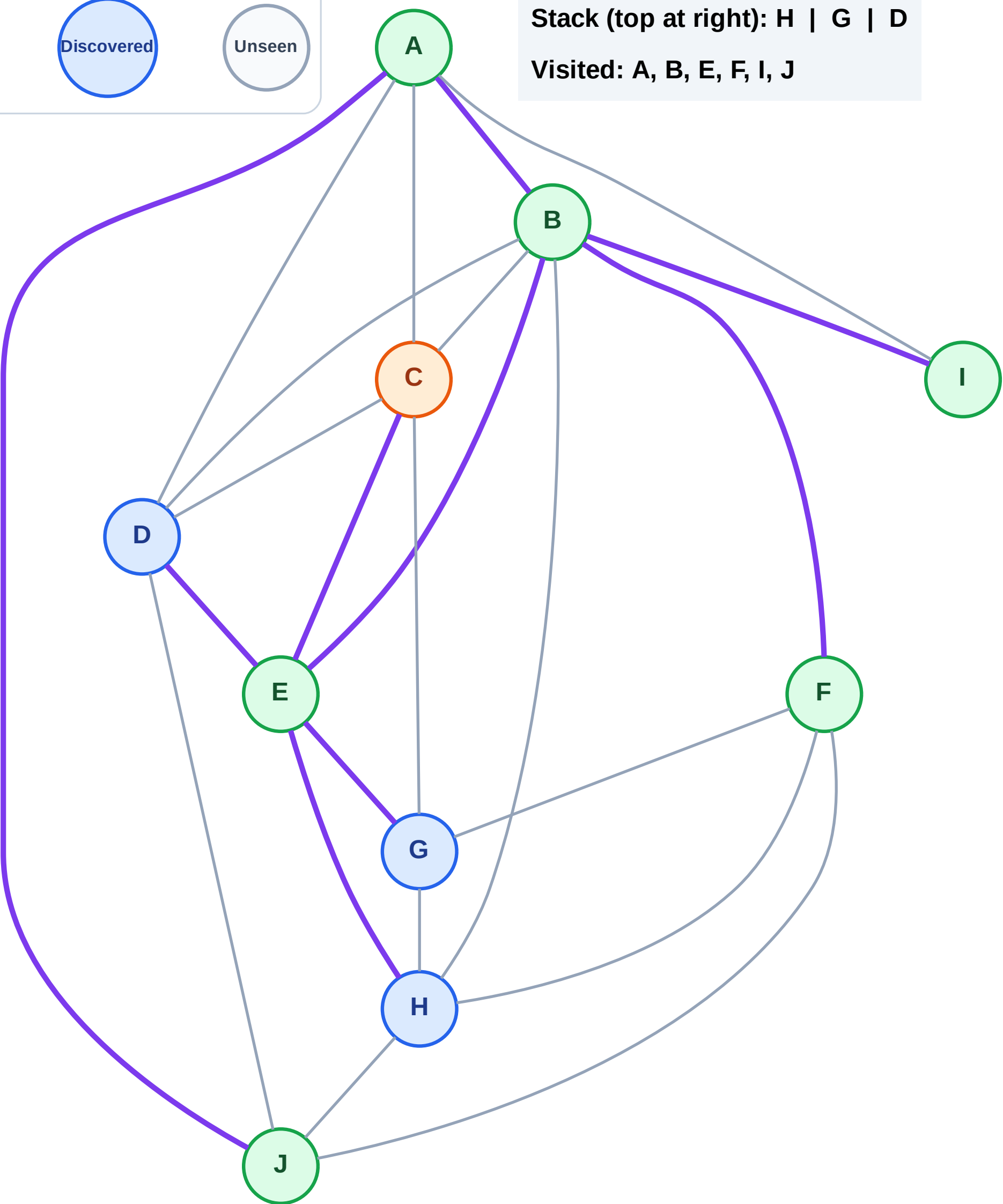
Processing

Discovered

Unseen

Stack (top at right): H | G | D

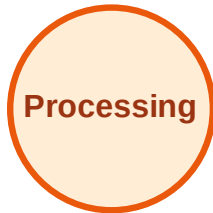
Visited: A, B, E, F, I, J



DFS Traversal

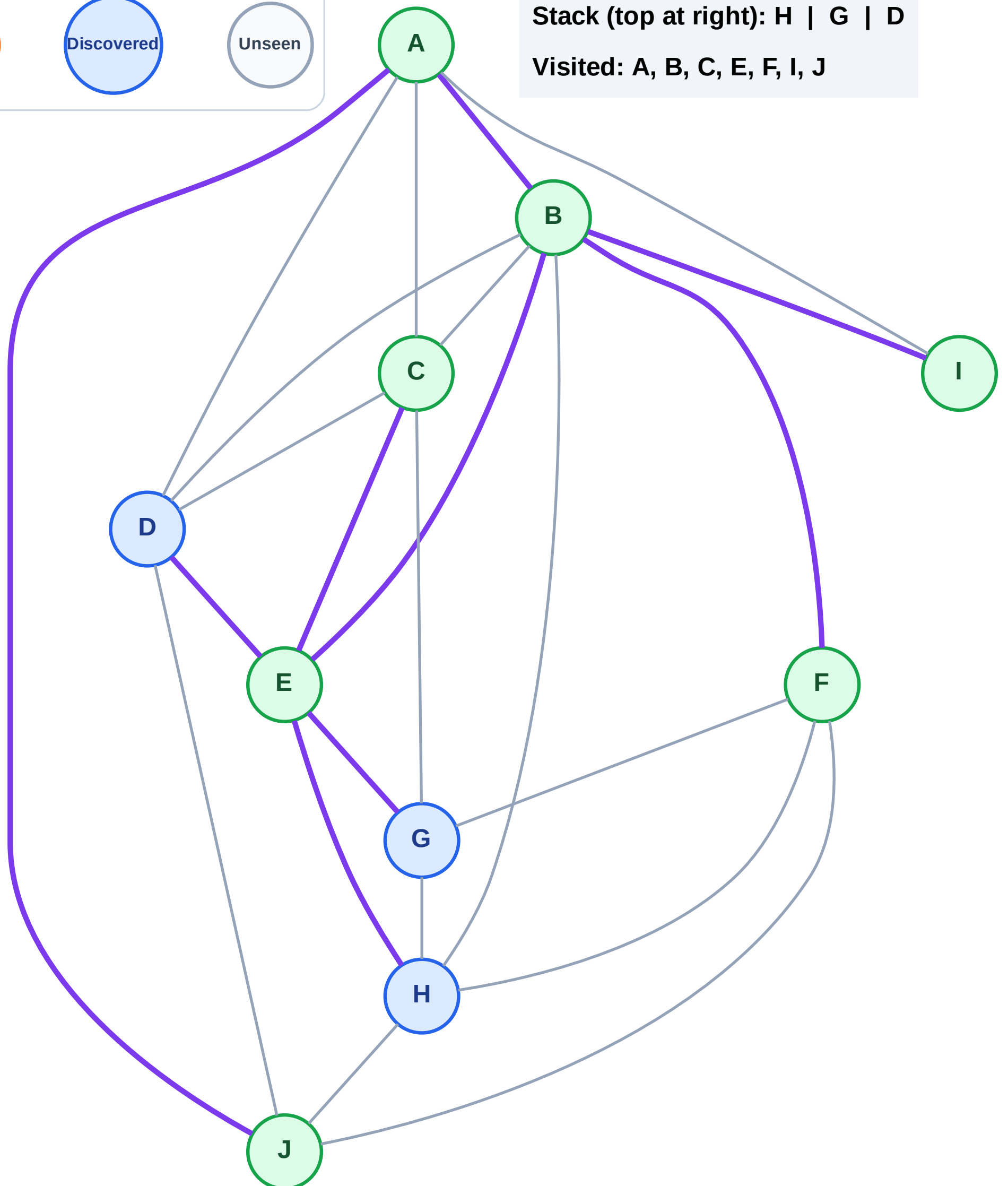
Step 15: No new neighbors from C

Legend



Stack (top at right): H | G | D

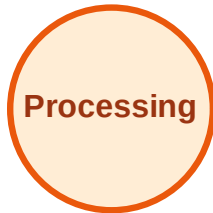
Visited: A, B, C, E, F, I, J



DFS Traversal

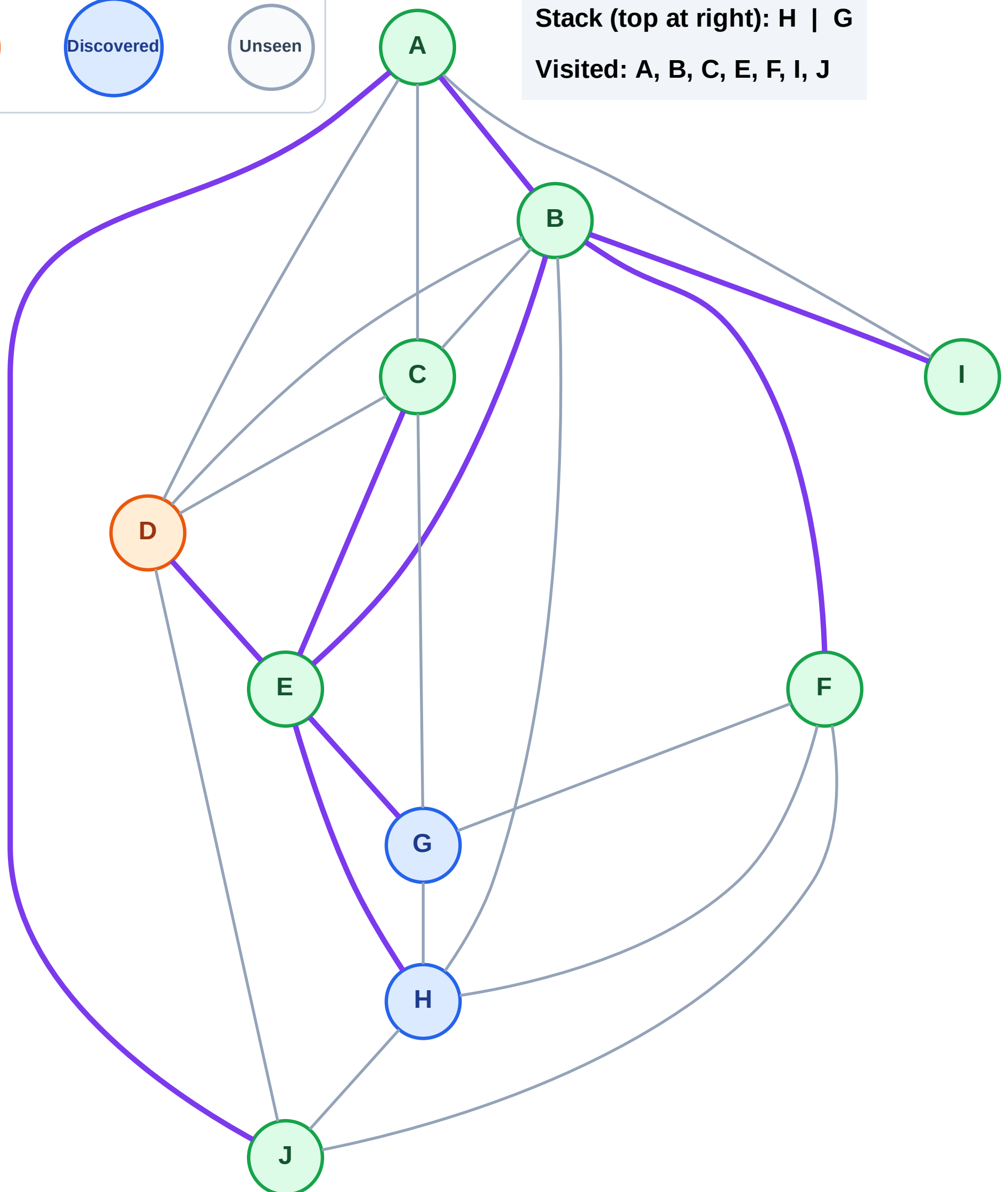
Step 16: Process node D

Legend



Stack (top at right): H | G

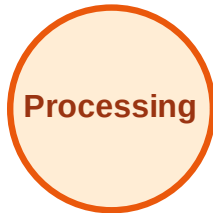
Visited: A, B, C, E, F, I, J



DFS Traversal

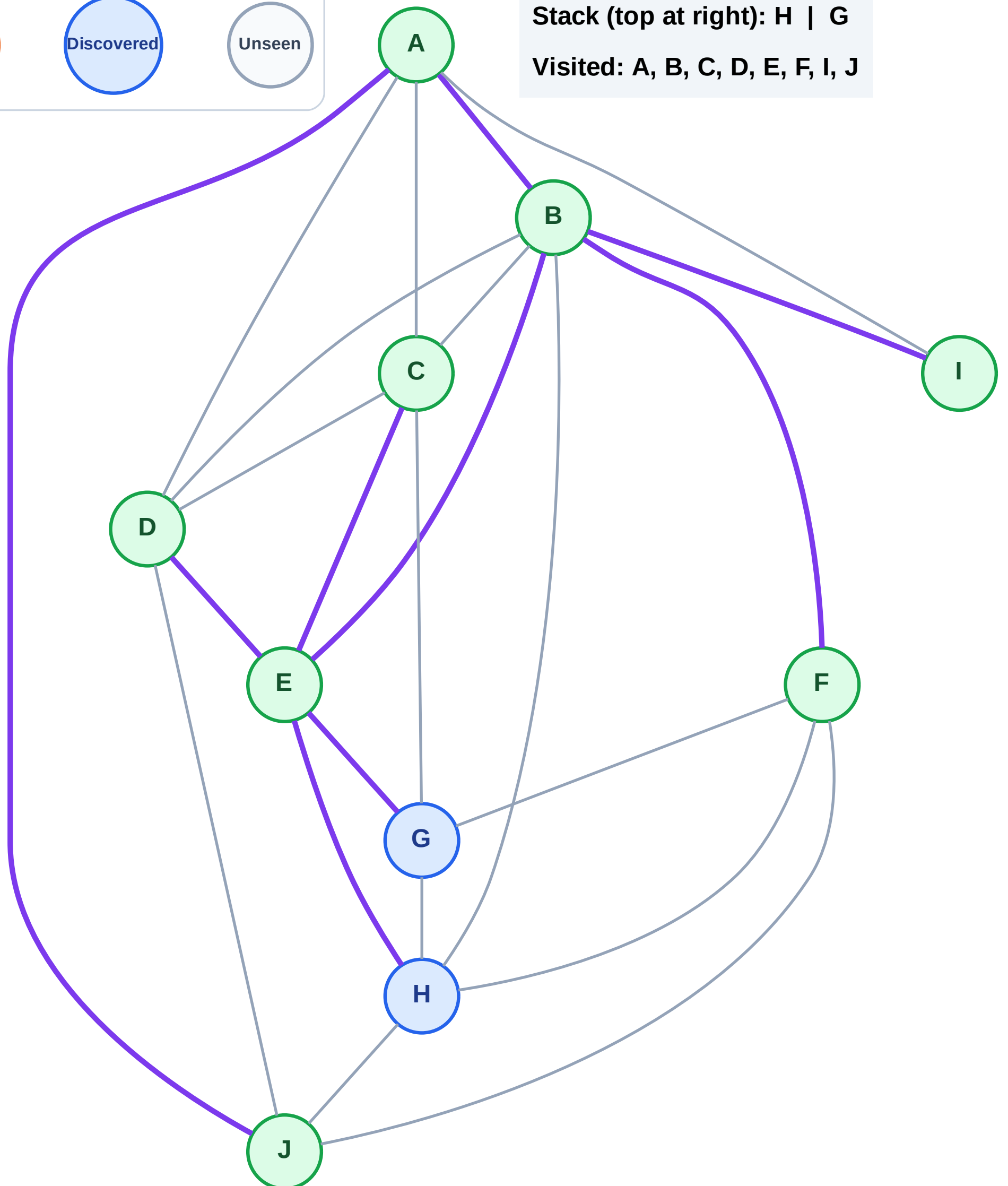
Step 17: No new neighbors from D

Legend



Stack (top at right): H | G

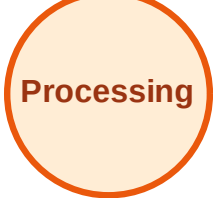
Visited: A, B, C, D, E, F, I, J



DFS Traversal

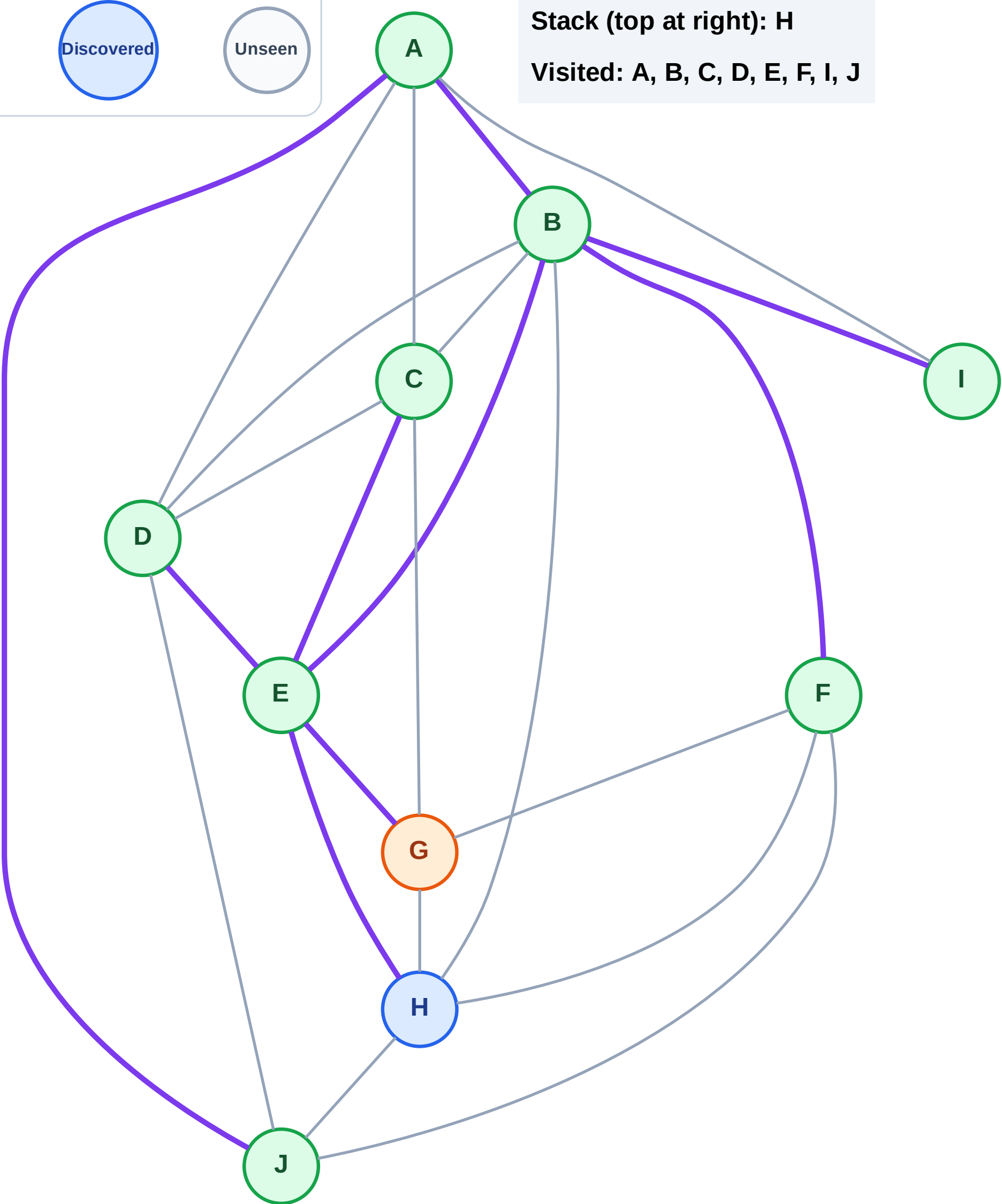
Step 18: Process node G

Legend



Stack (top at right): H

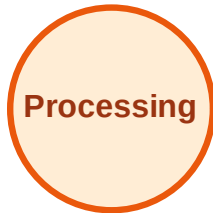
Visited: A, B, C, D, E, F, I, J



DFS Traversal

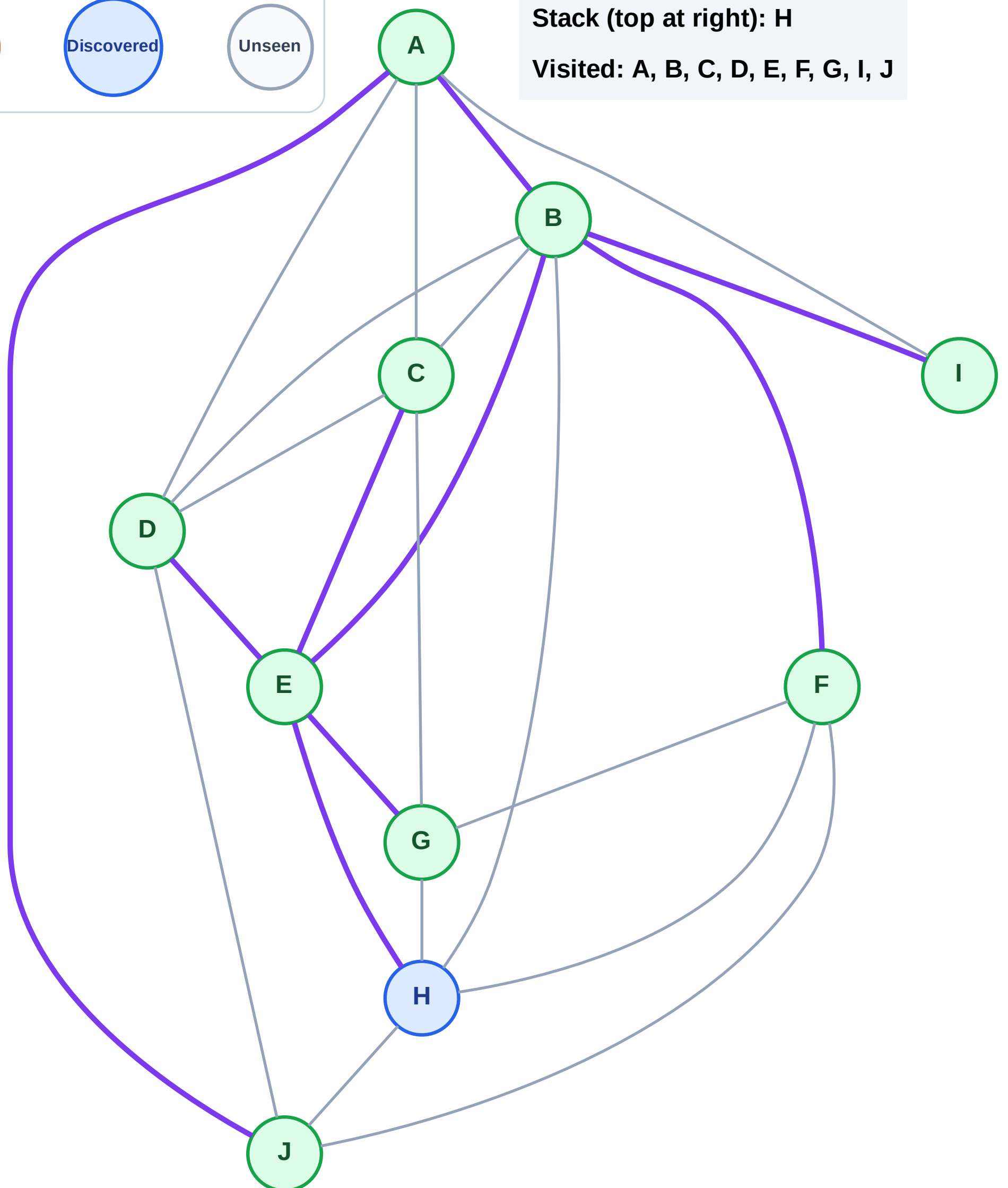
Step 19: No new neighbors from G

Legend



Stack (top at right): H

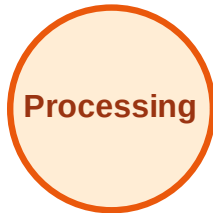
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

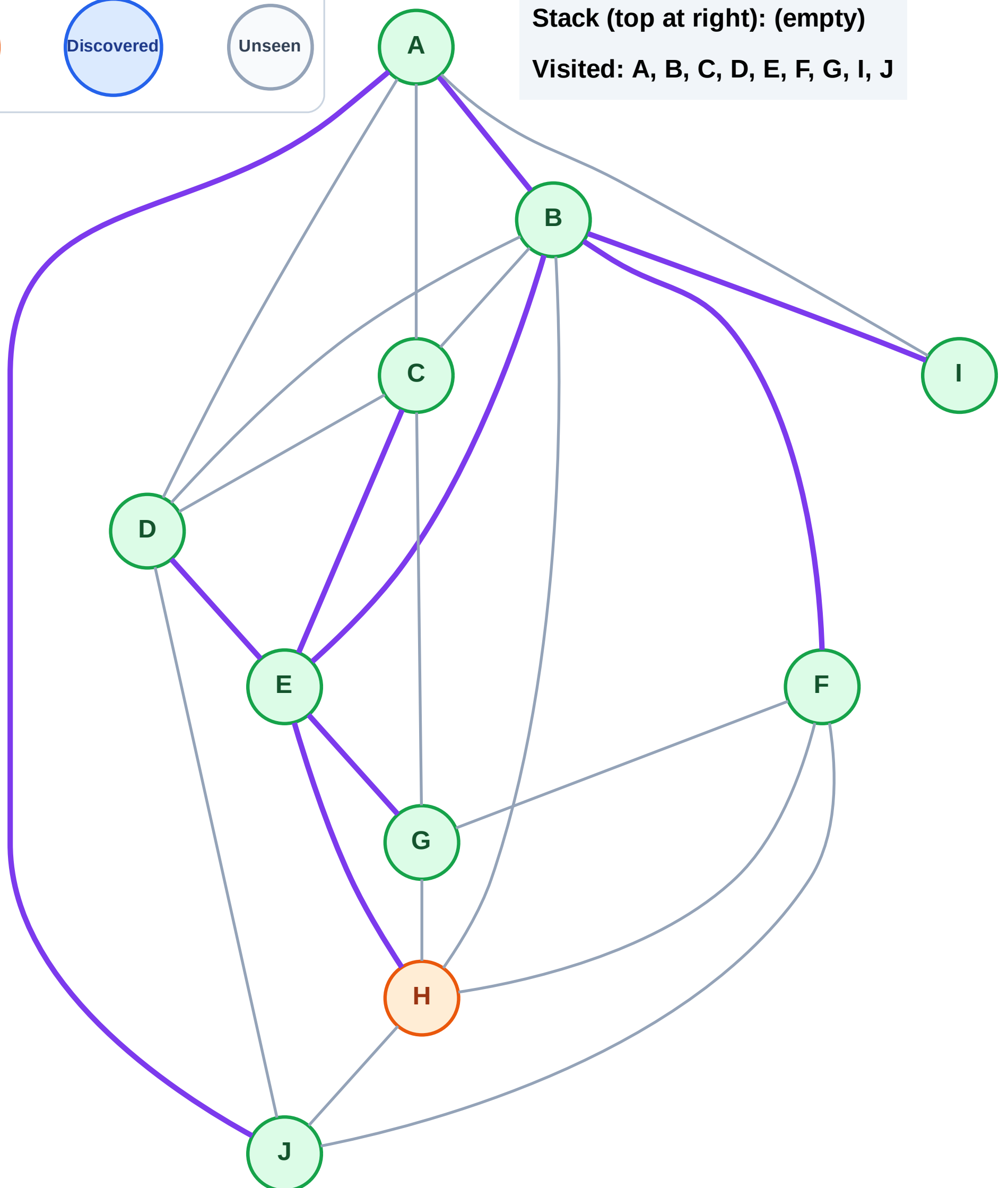
Step 20: Process node H

Legend



Stack (top at right): (empty)

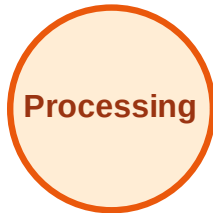
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

